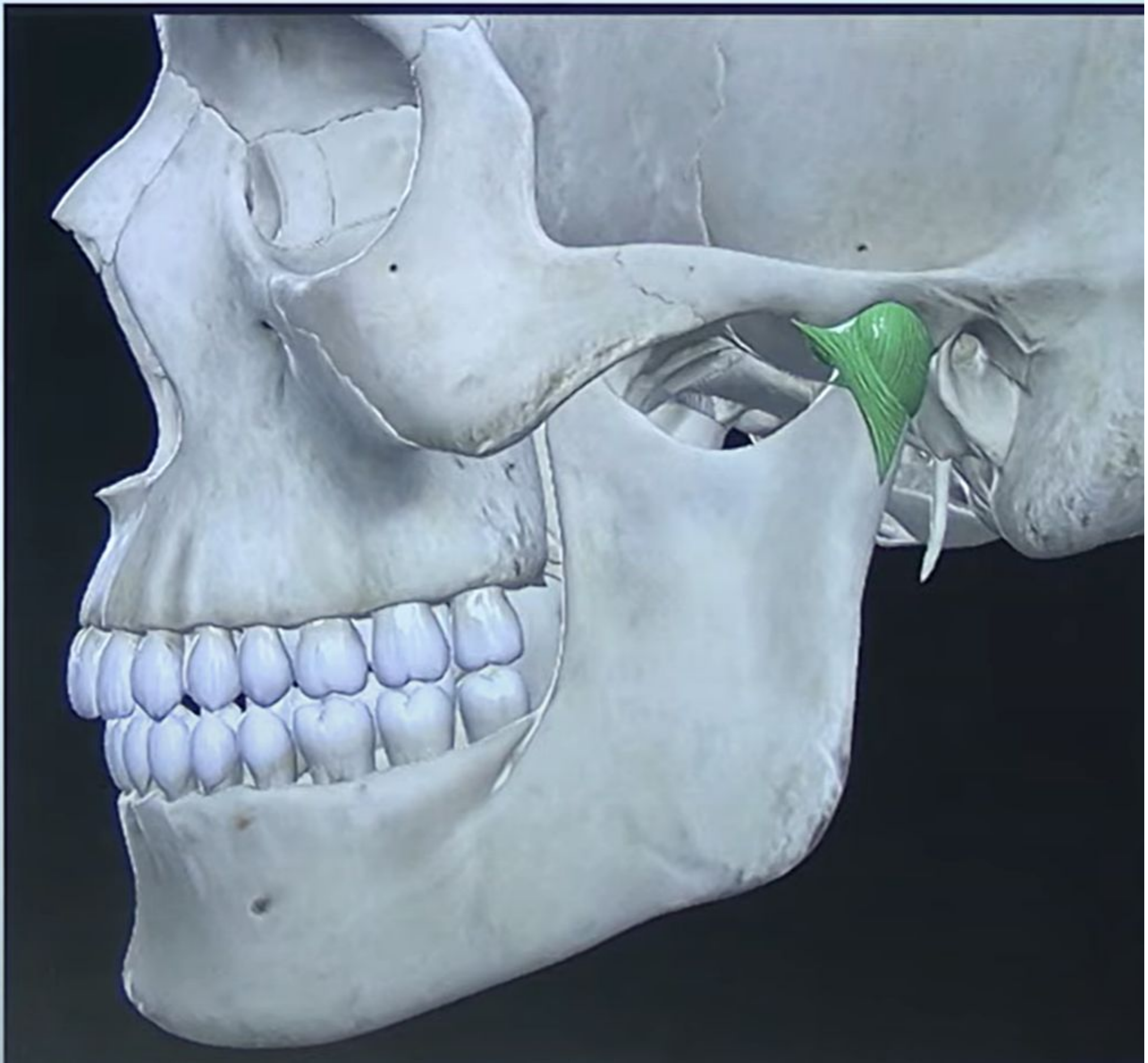

ANATOMY OF TEMPORO-MAN DIBULAR JOINT

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Learning Objectives

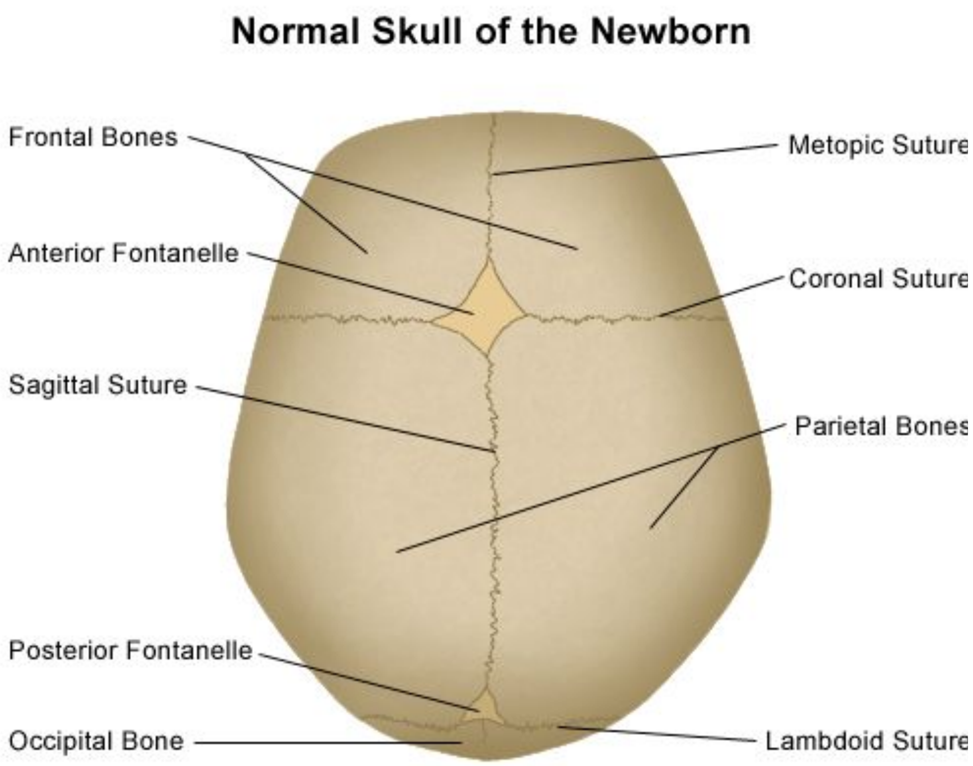
- Understand the **anatomy of TMJ**
- Identify **mandibular fossa, condyle, ligaments, and disc**
- Explain **hinge and translational movements**
- Describe **innervation, blood supply, and muscles of mastication**
- Relate **clinical aspects of TMJ disorders**

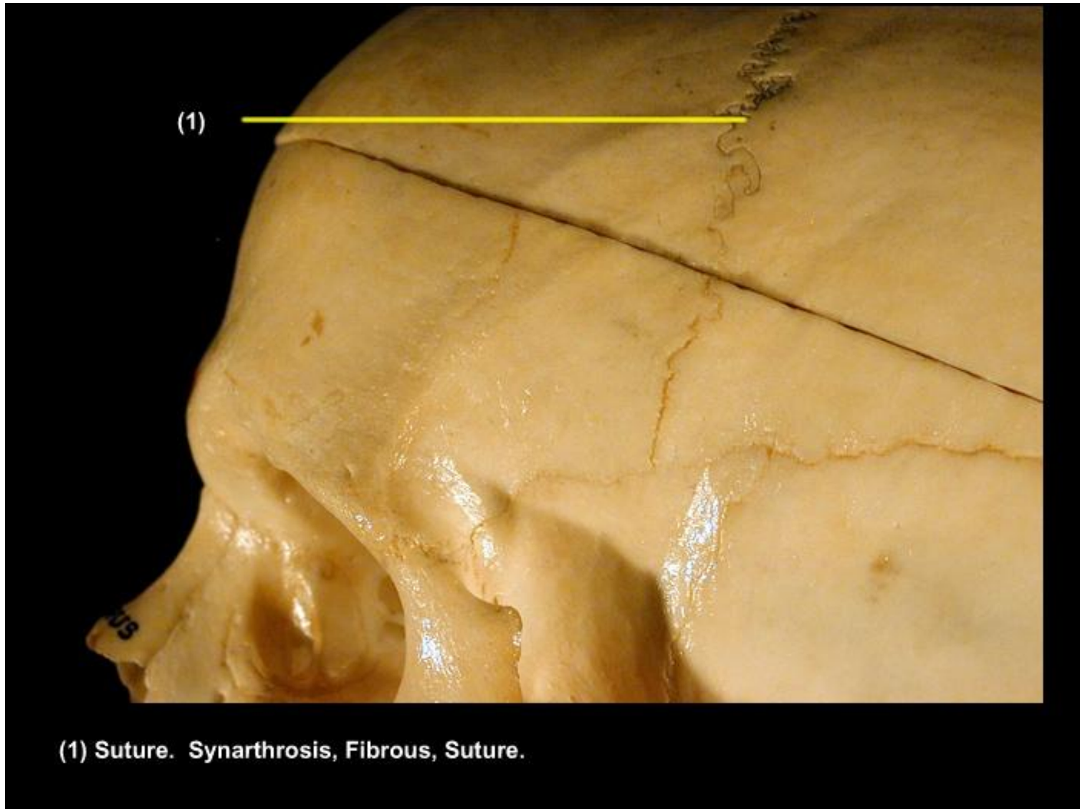
CLASSIFICATION OF JOINTS

1. FIBROUS JOINTS: Two bones are connected by fibrous tissue.

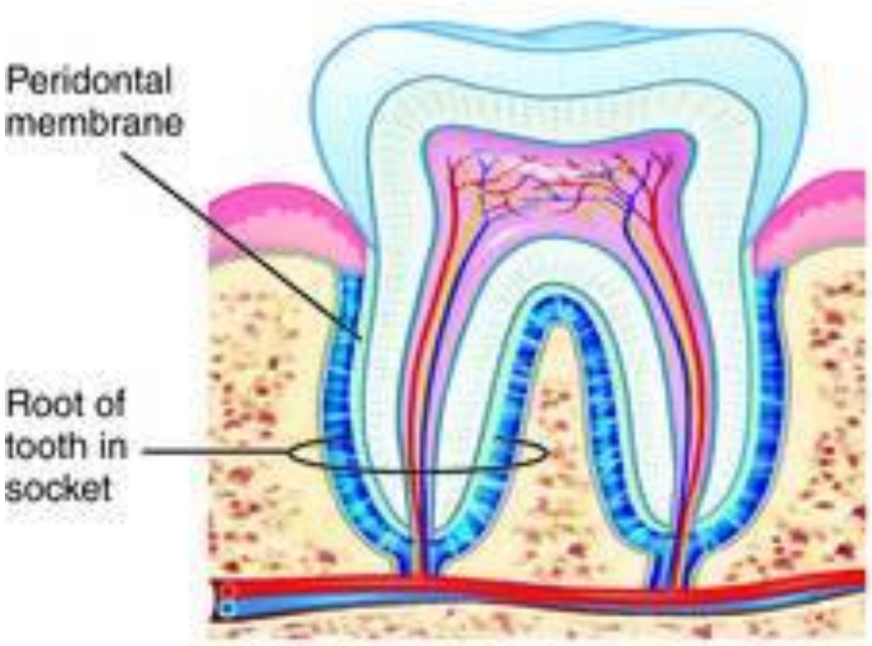
Three types:

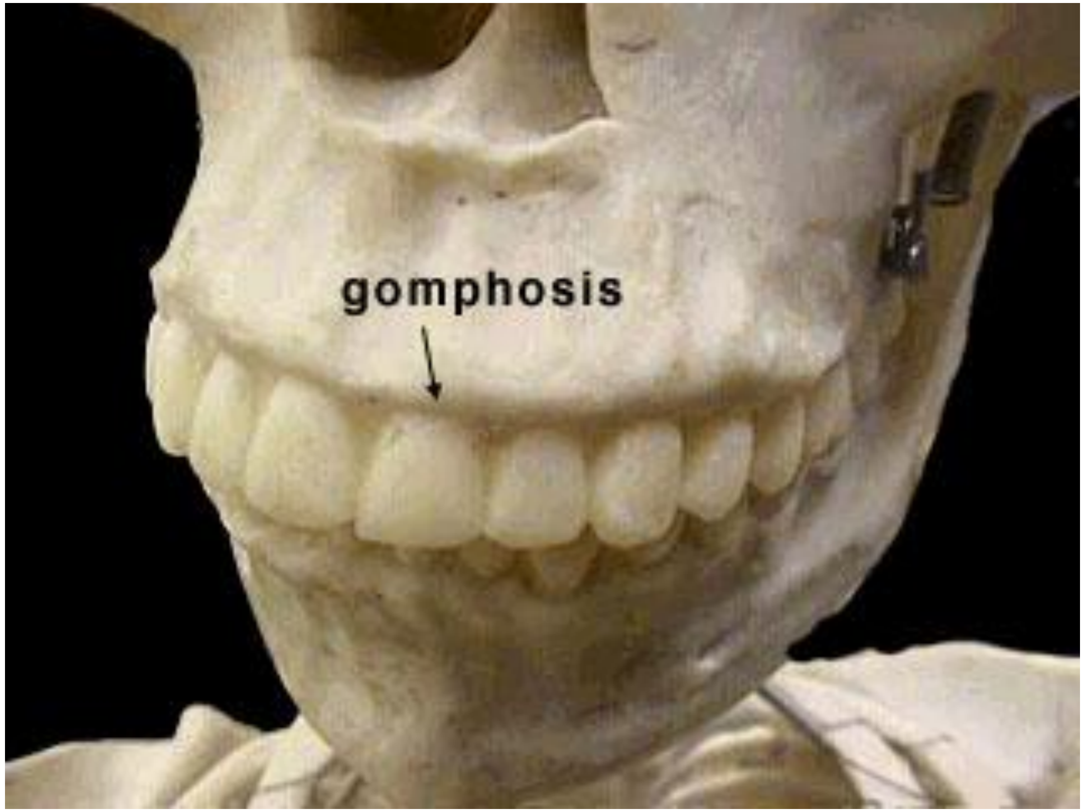
- Suture
- Gomphosis
- syndesmosis


Normal Skull of the Newborn
Frontal Bones
Anterior Fontanelle
Sagittal Suture
Posterior Fontanelle
Occipital Bone
Metopic Suture
Coronal Suture
Parietal Bones
Lambdoid Suture

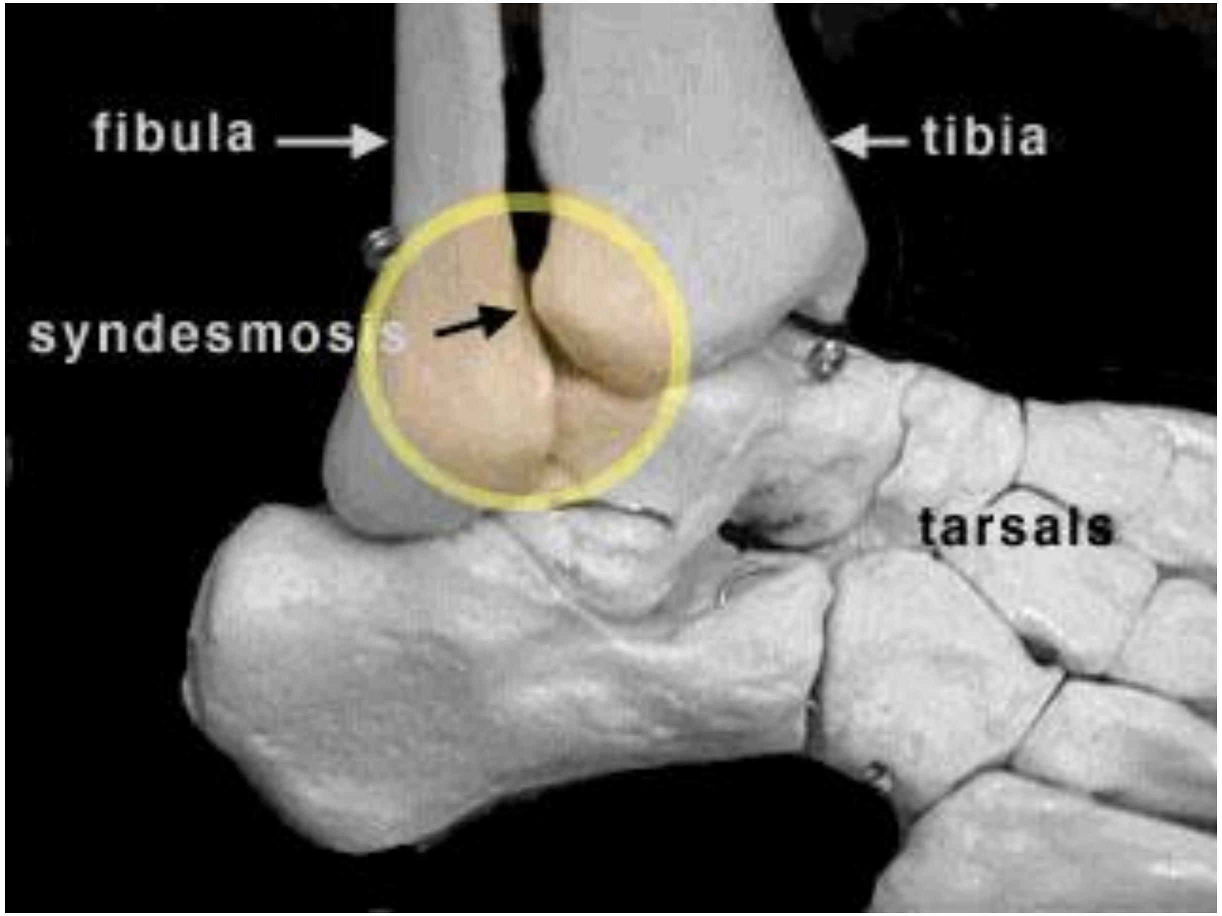

(1)
(1) Suture. Synarthrosis, Fibrous, Suture.

1a. SUTURE





1b. GOMPHOSIS

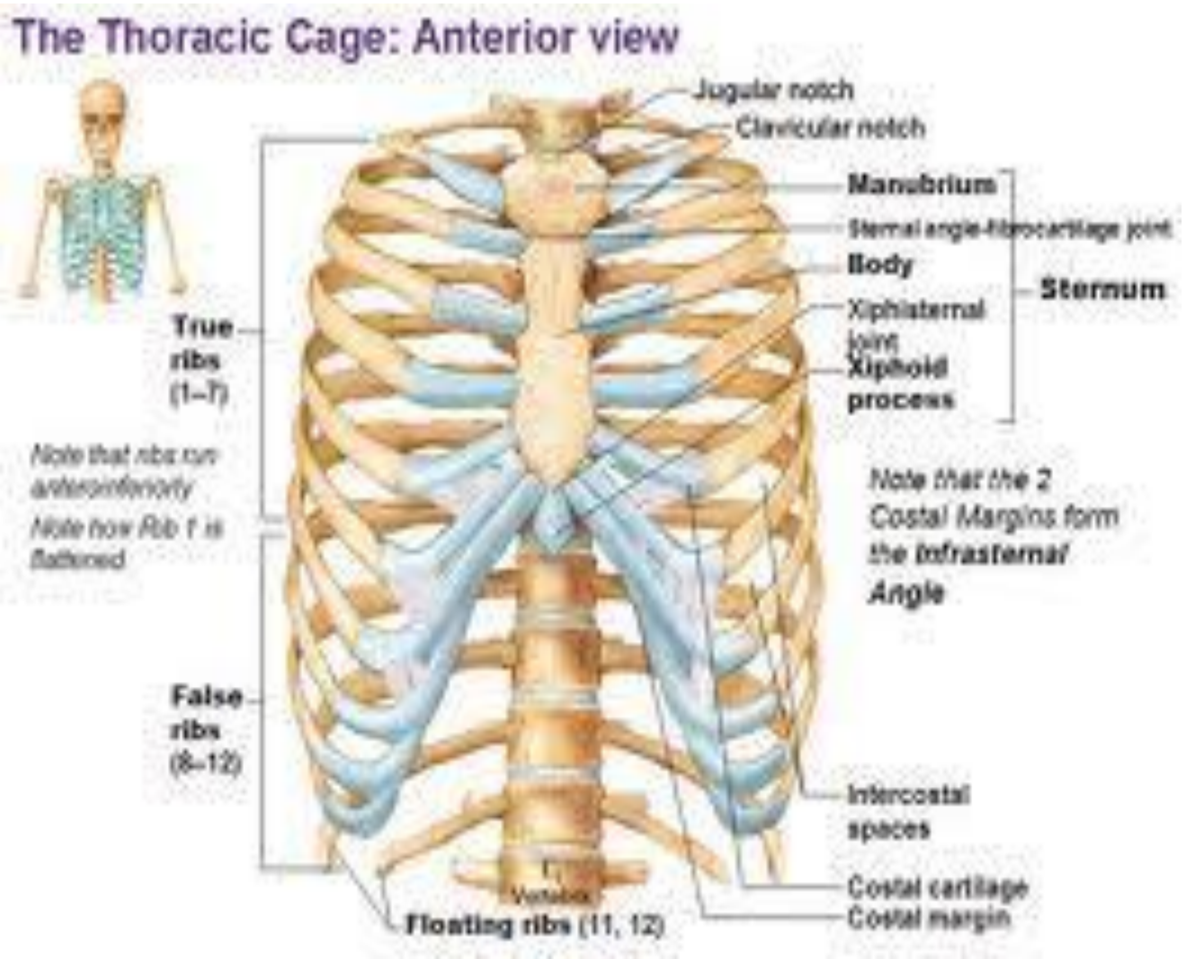


1c. SYNDESMOSIS

CARTILAGINOUS JOINTS

Primary Cartilaginous Joints:

Direct Apposition of Bone & Cartilage



e.g. Costochondral Junction.

CARTILAGINOUS JOINTS

Secondary cartilaginous joints: articulation occur in the sequence as bone-cartilage-fibrous tissue-cartilage-bone. e.g. pubic symphysis.

Cartilaginous & fibrous joints permit little if any movement b/w the bones involved.

The temporomandibular joints

- one on each side, allow opening and closing of the mouth and complex chewing or side-to-side movements of the lower jaw.

- **Type of Joint:** This is a synovial joint of the condylar variety.

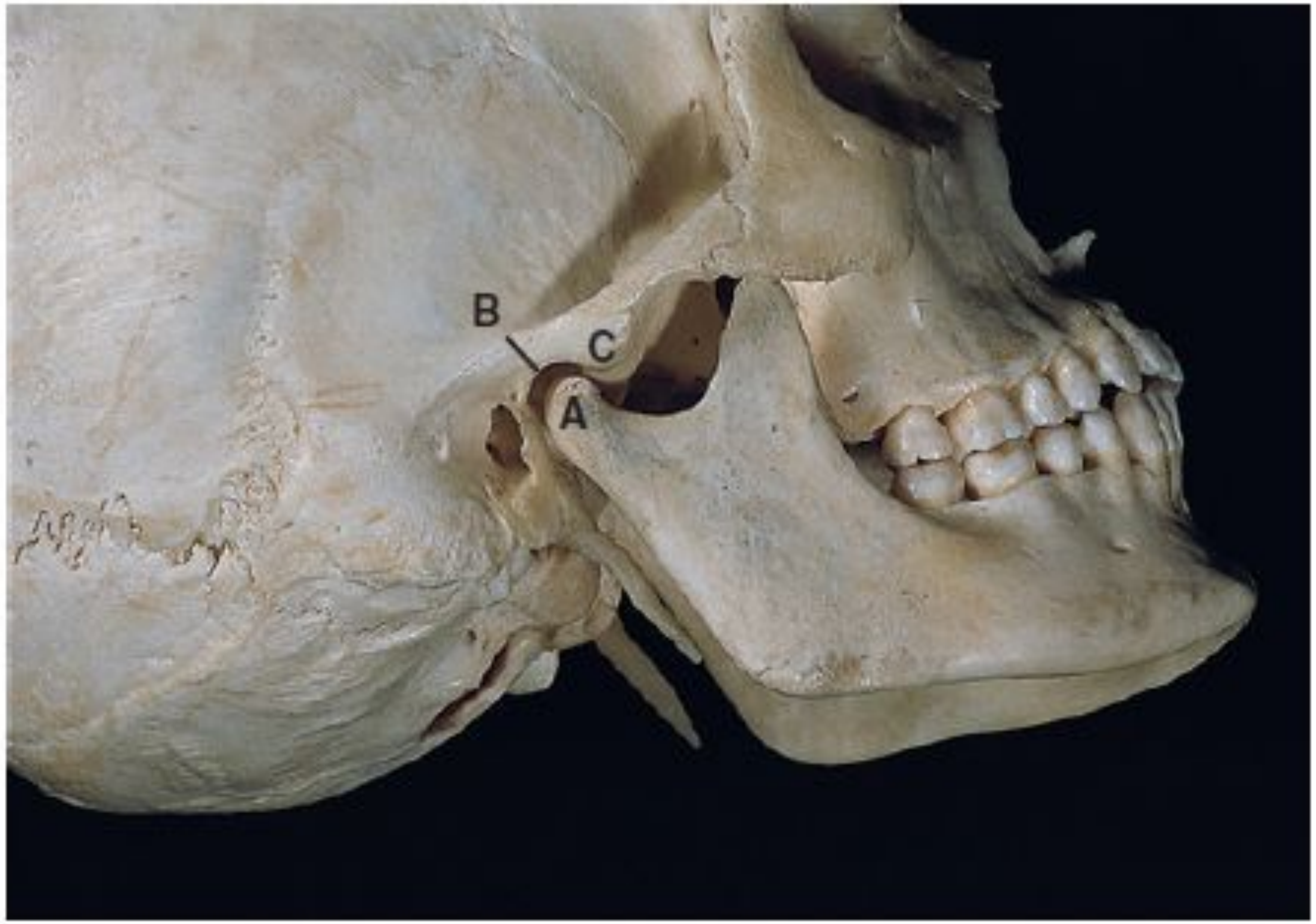


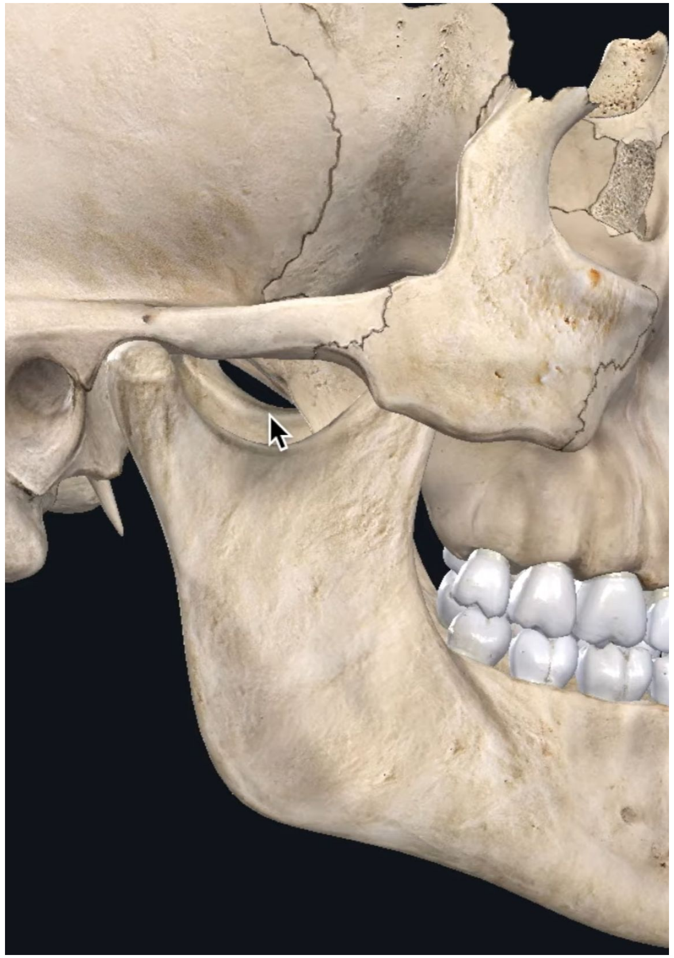
Fig. 3.1 The osteology of the temporomandibular joint. A = mandibular condyle; B = mandibular fossa of temporal bone; C = articular eminence of temporal bone.

TEMPOROMANDIBULAR JOINT

SYNOVIAL JOINT

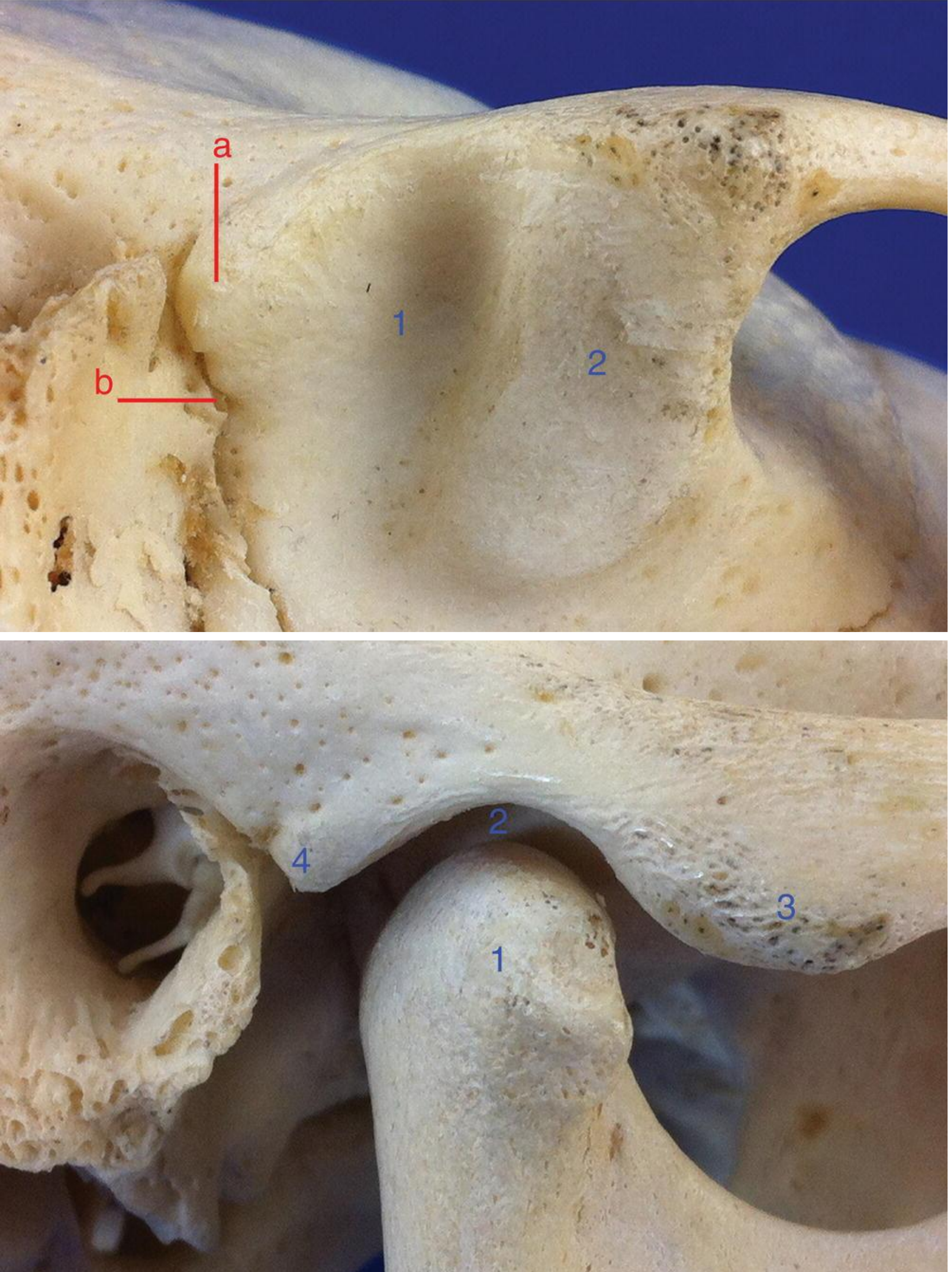
- translatory and rotatory movements occurs; so also called as synovial sliding-ginglymoid (hinge) joint.

TEMPOROMANDIBULAR JOINT



- **ARTICULAR SURFACES**
- Each joint formed between the head of the mandible and the articular fossa and articular tubercle of the temporal bone.
- The upper articular bone is immobile, while lower bone alone is mobile.

Permits significant movement.



ARTICULAR SURFACES

- upper articular surface is concavo-convex.
- It is formed by the following parts of the temporal bone:
  - 3# Articular tubercle of zygomatic process.
  - 1# Anterior part of mandibular fossa of squamous temporal bone.
  - Posterior non-articular part behind the squamotympanic fissure formed by the tympanic plate (triangular).

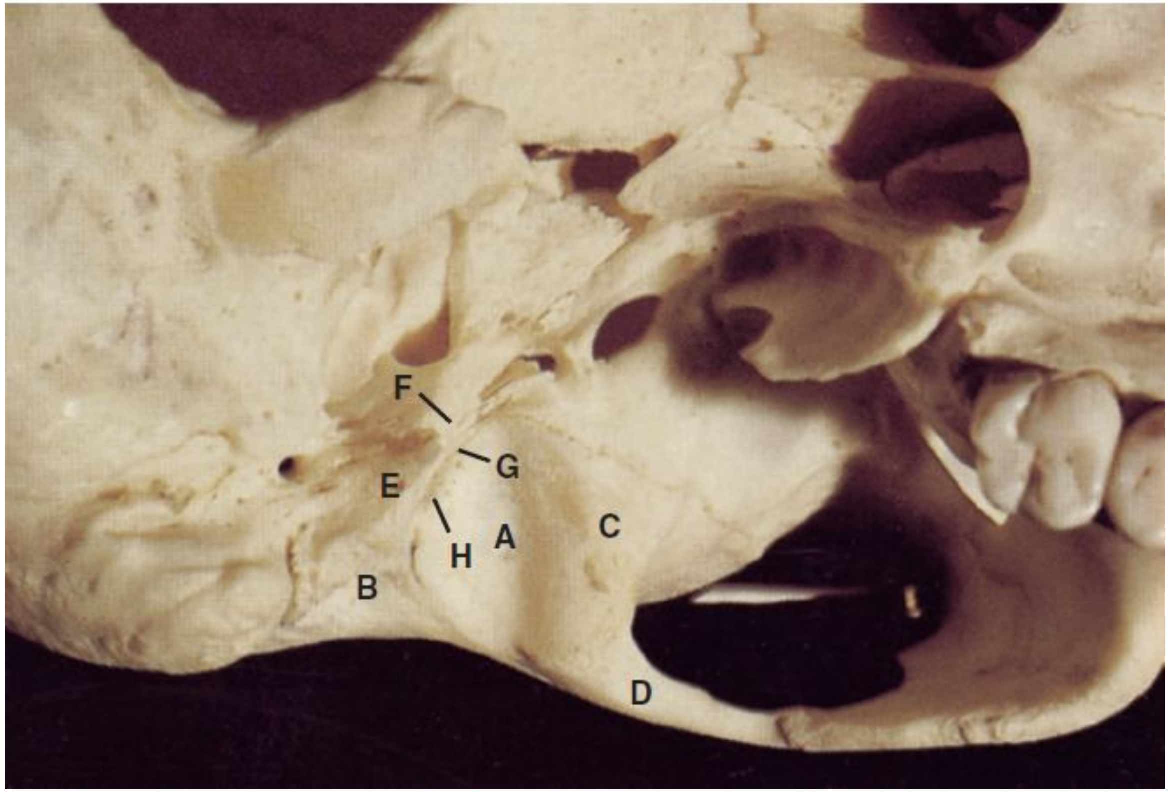


Fig. 3.2 The osteology of the mandibular fossa. A = mandibular fossa; B = external acoustic meatus; C = articular eminence; D = zygomatic process of temporal bone; E = tympanic plate; F = petrotympanic fissure; G = petrosquamous fissure; H = squamotympanic fissure.

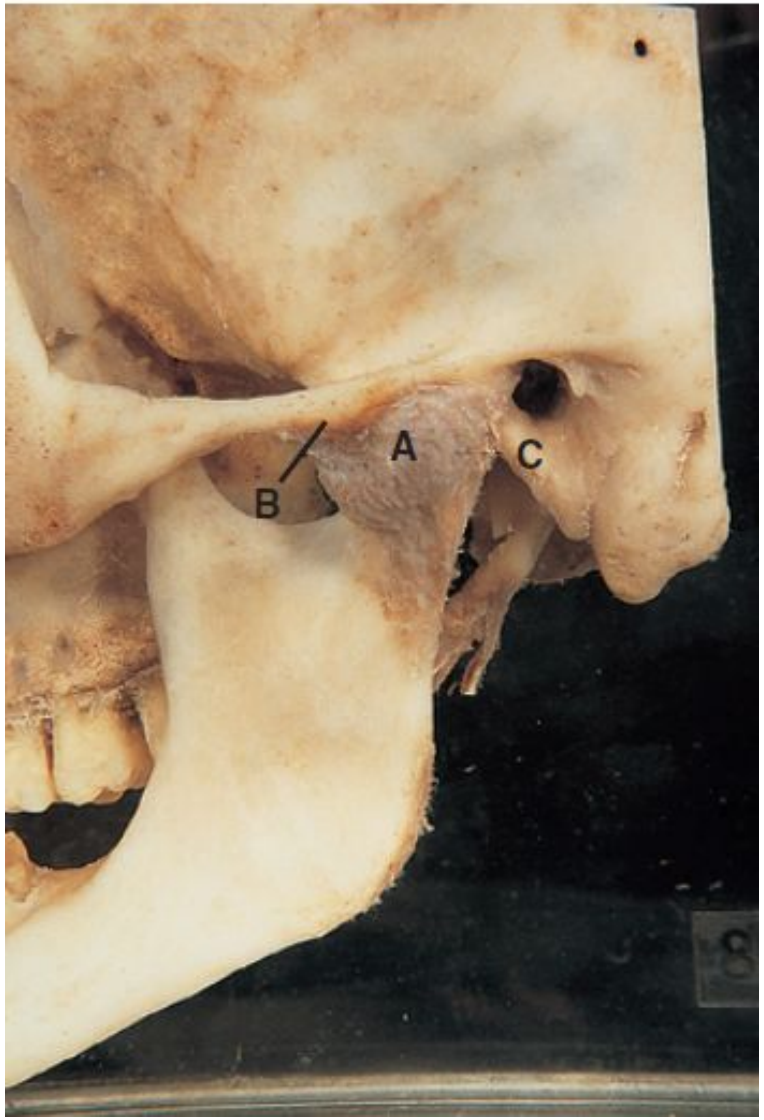
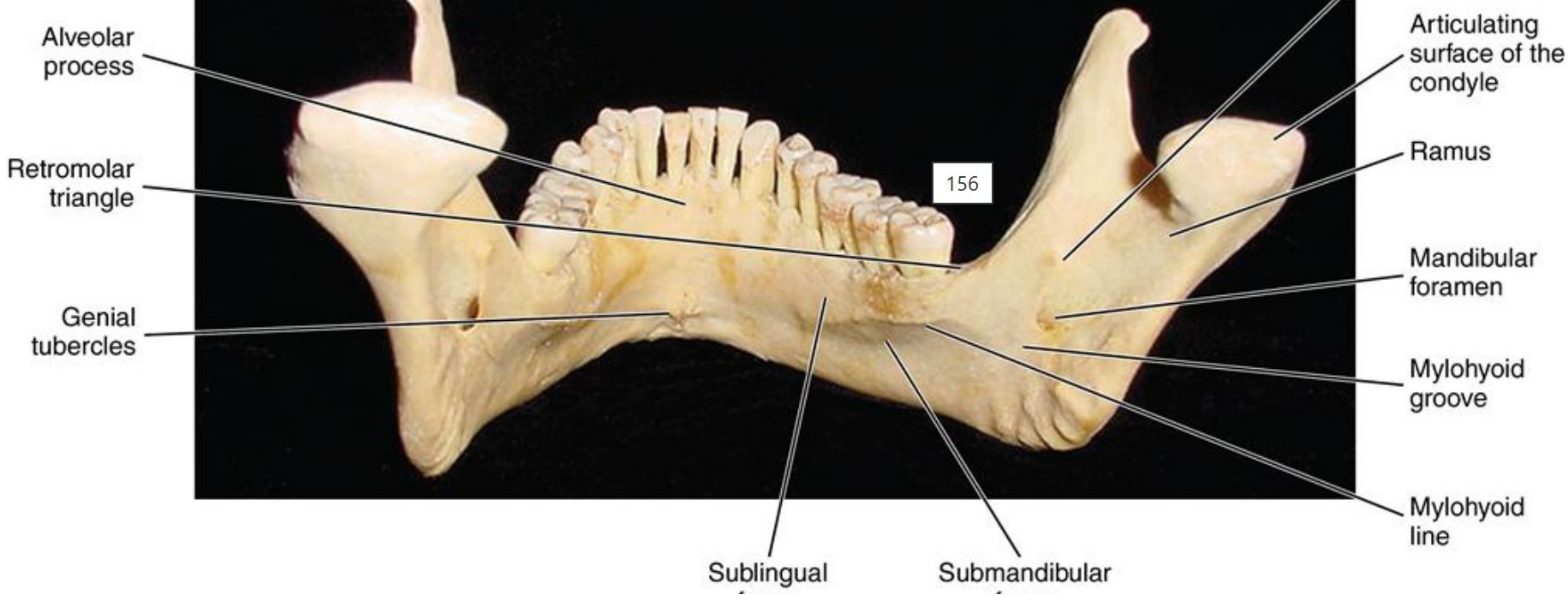


Fig. 3.5 The capsule of the temporomandibular joint (A). B = articular eminence; C = tympanic plate.

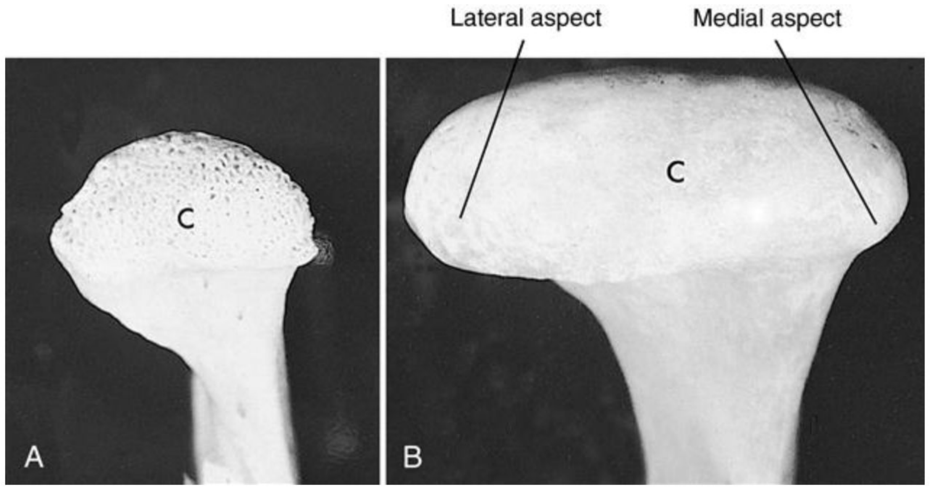


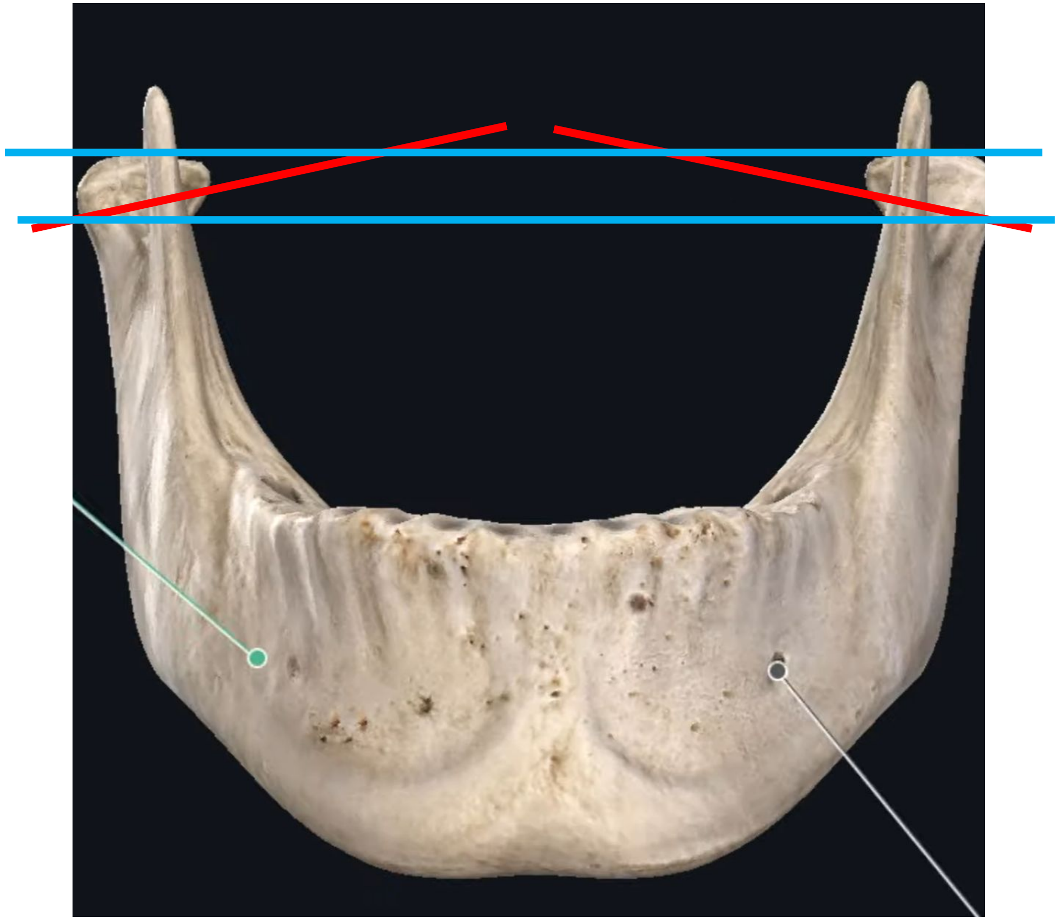
Inferior Articular Surface

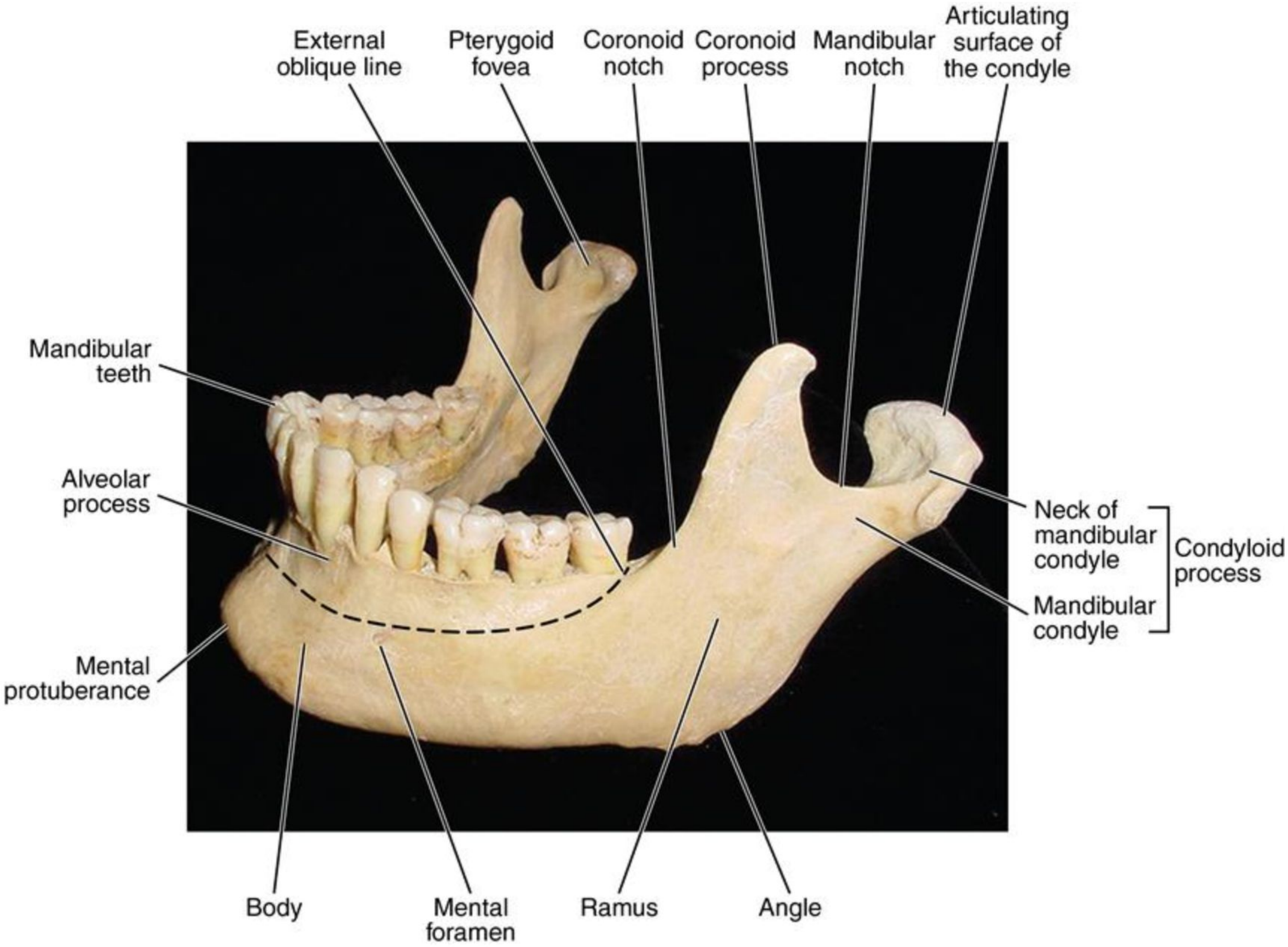
- The inferior articular surface is formed by the head of the mandible which are convex.
- Each condyle of mandible is elliptical.

Inferior Articular Surface

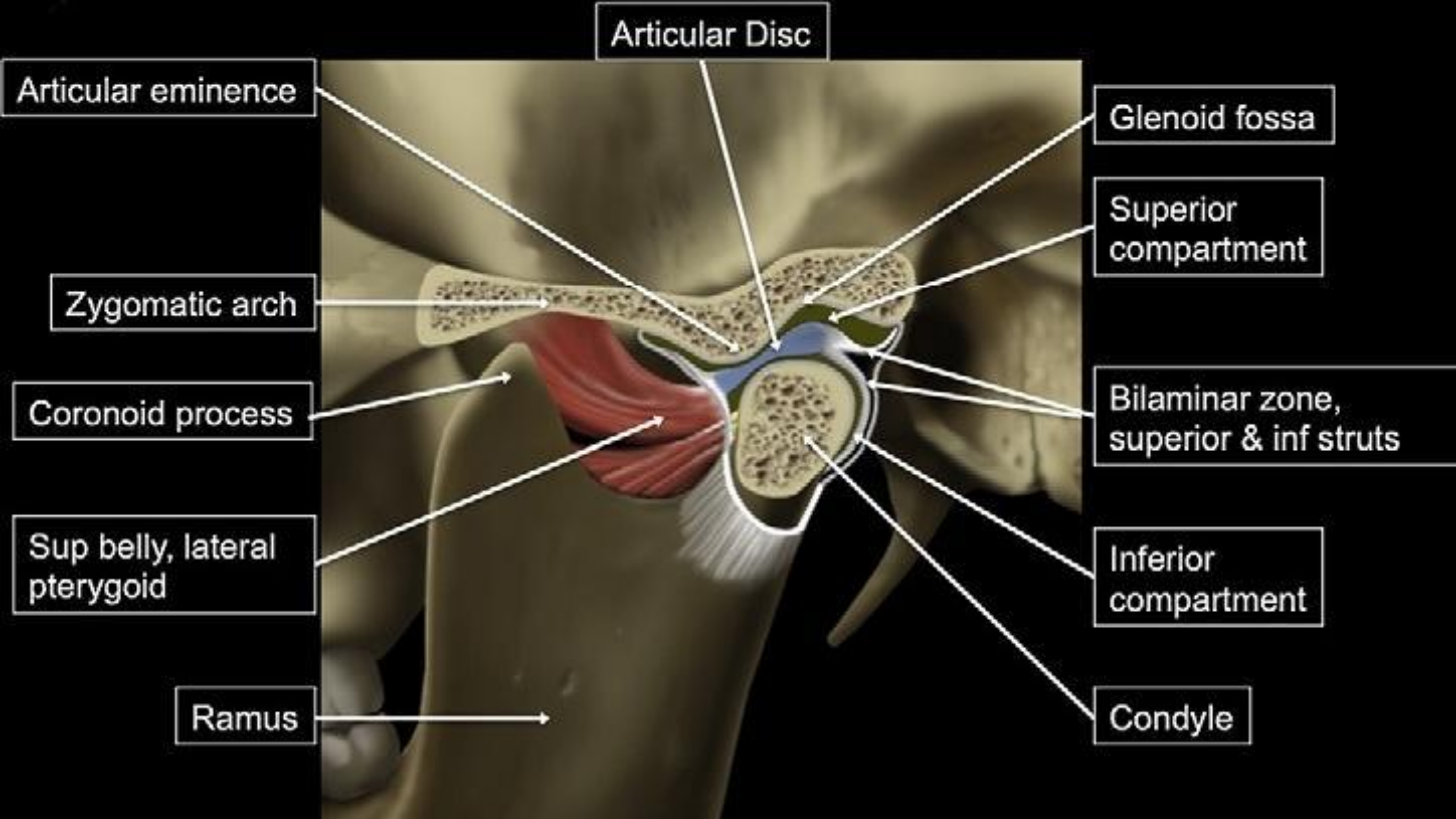
- The long axis of condyles oriented medio-laterally, directed backwards and medially.
- These both sides axis form an arc of circle, which passes through the anterior margin of the foramen magnum.







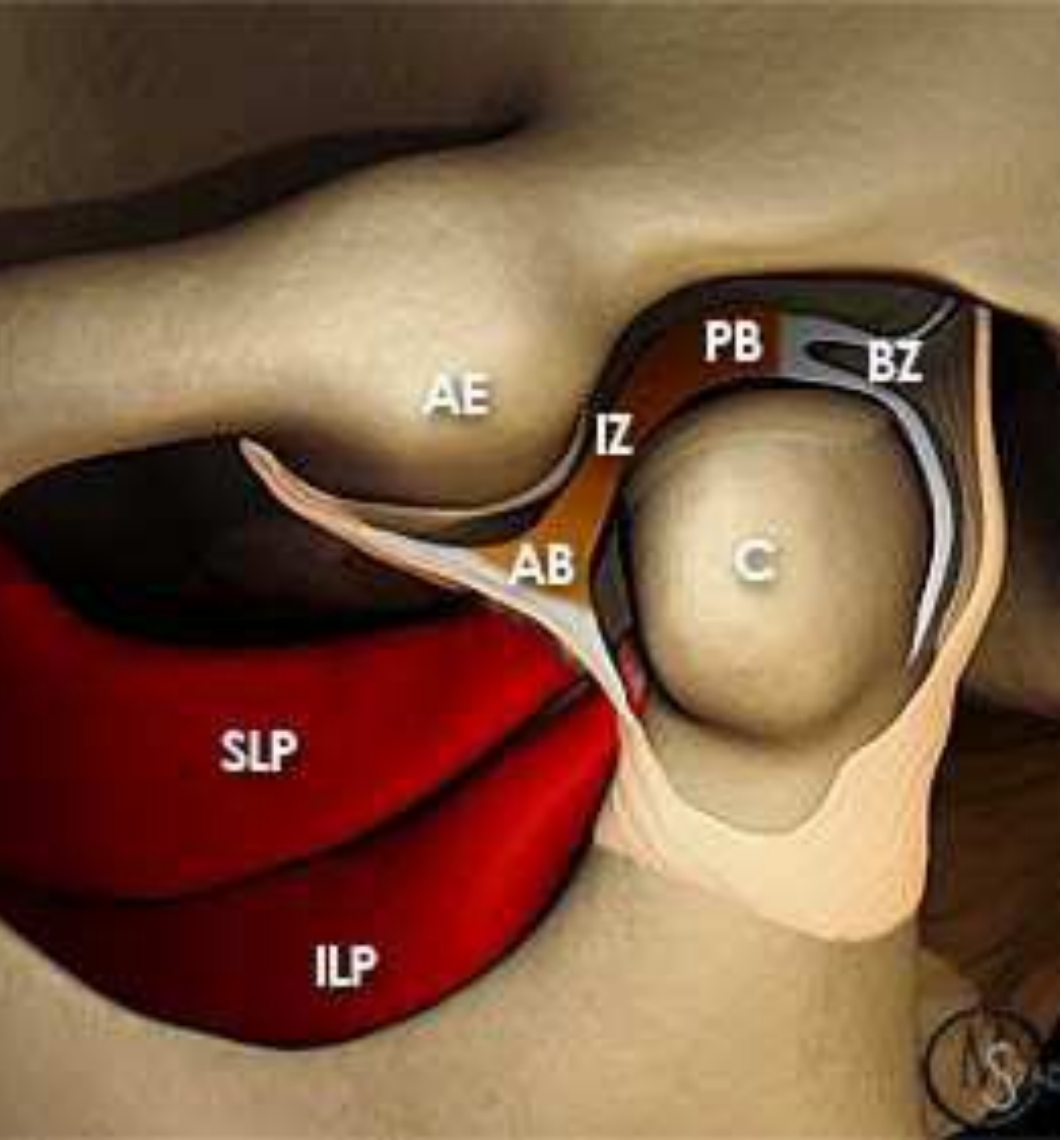
External oblique line
Pterygoid fovea
Coronoid notch
Coronoid process
Mandibular notch
Articulating surface of the condyle
Mandibular teeth
Alveolar process
Neck of mandibular condyle
Condyloid process
Mandibular condyle
Mental protuberance
Body
Mental foramen
Ramus
Angle


Articular Disc
Articular eminence
Glenoid fossa
Superior compartment
Zygomatic arch
Coronoid process
Bilaminar zone, superior & inf struts
Sup belly, lateral pterygoid
Inferior compartment
Ramus
Condyle

- Unlike most other synovial joints where the articular surfaces of the bones are covered by a layer of hyaline cartilage, those of the temporomandibular joint are covered by **fibrocartilage.**
- The joint cavity is completely divided into upper and lower parts by an **intra-articular disc.**

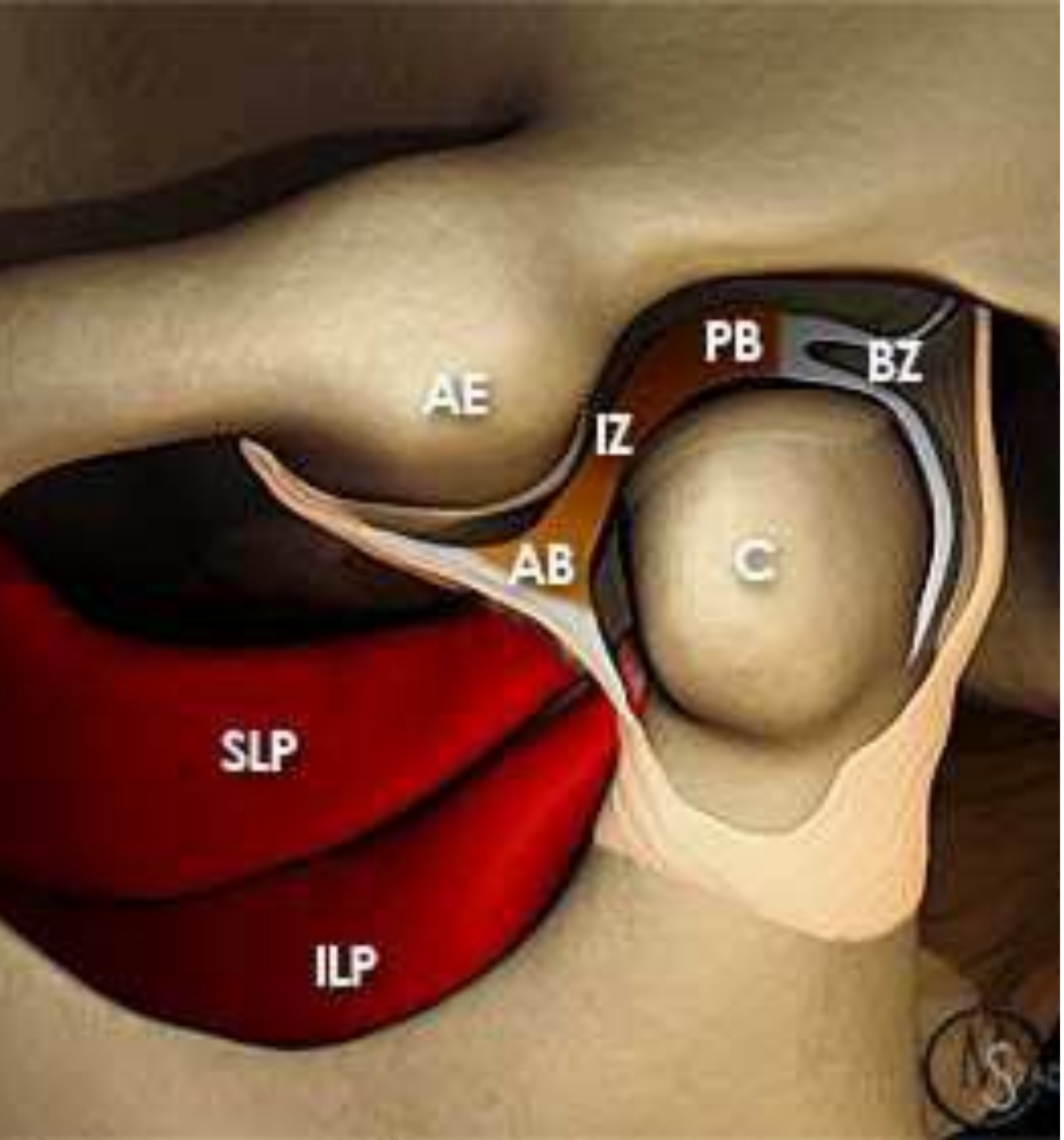
1. Upper menisco-temporal compartment- translational movement
2. Lower menisco-mandibular compartment- rotational movement

- **Fibrocartilage = “FIBER + FORCE”** → strong, tough
- **Hyaline cartilage = “HYALINE = GLASSY”** → smooth, slippery



SYNOVIAL JOINTS

- Cavity is filled with synovial fluid formed by a synovial membrane that lines the nonarticular surfaces.
- Various ligaments are associated to strengthen the articulation & check excess movements.



Articular eminence (AE)
Mandibular condyle (C)
Articular disc demonstrates a thicker anterior band (AB) and posterior band (PB) separated by a thinner intermediate zone (IZ). The bilaminar zone (BZ) and the superior (SLP) and inferior (ILP) bellies of the lateral pterygoid muscle are also indicated.

SYNOVIAL JOINTS

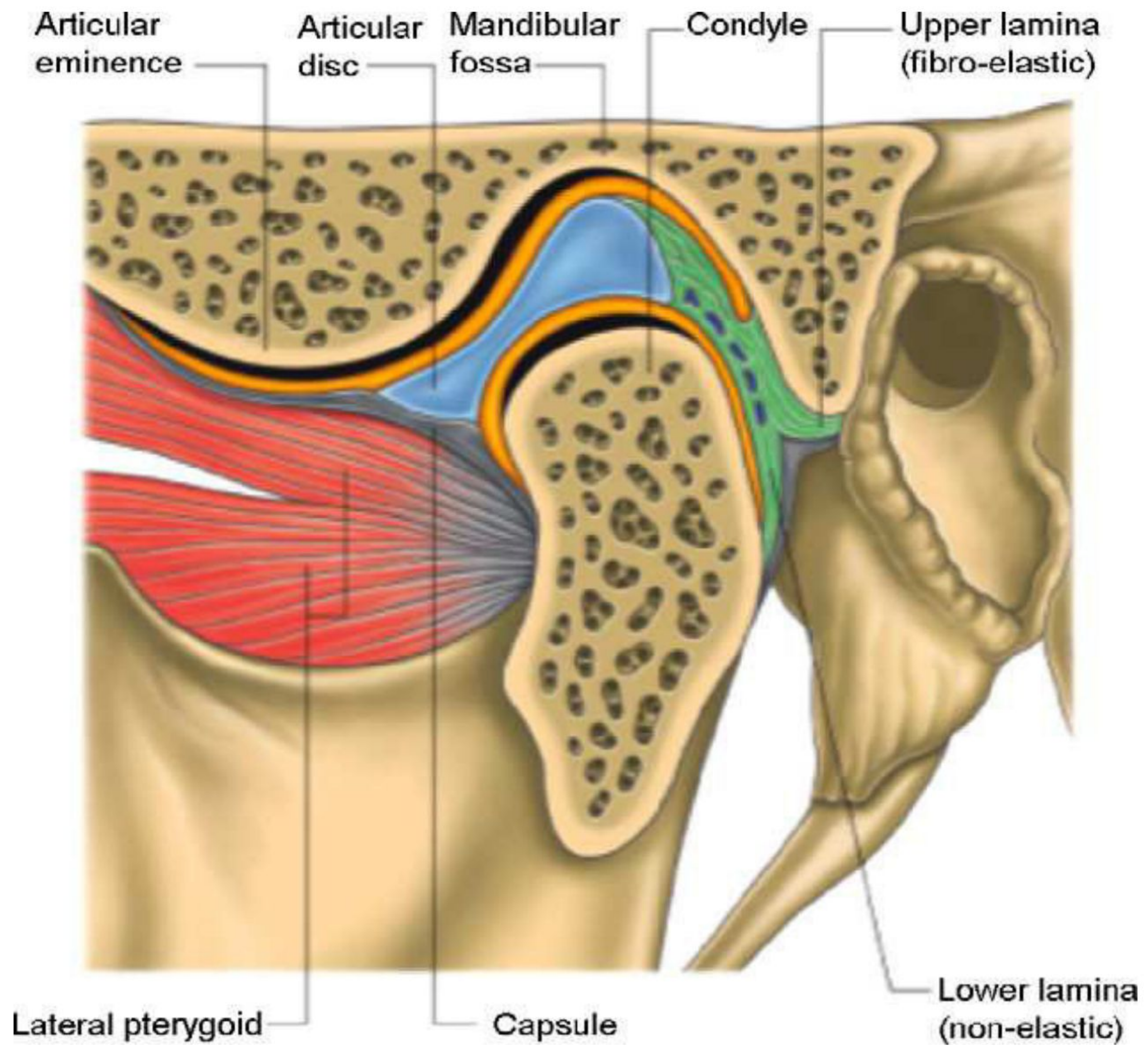
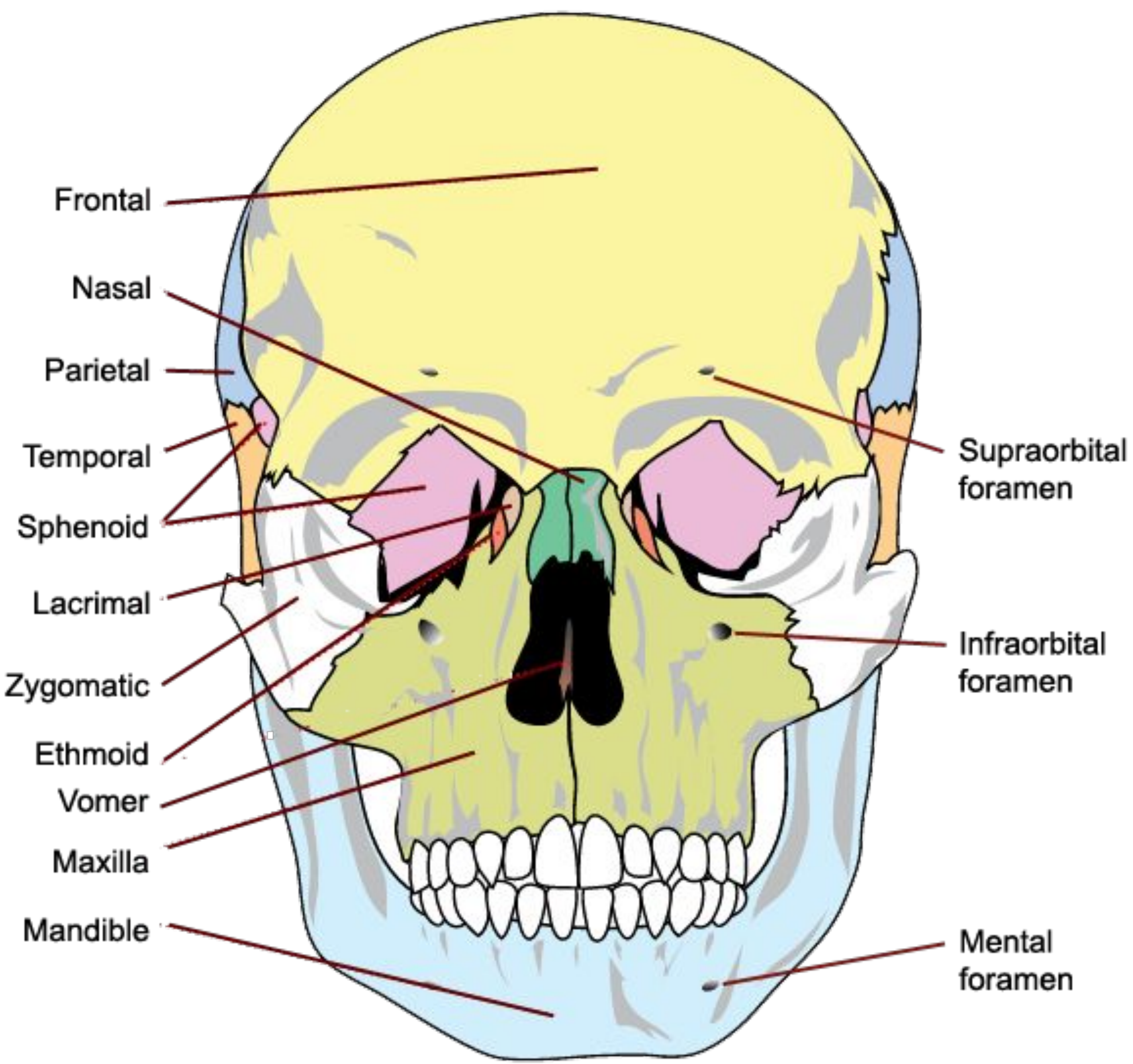
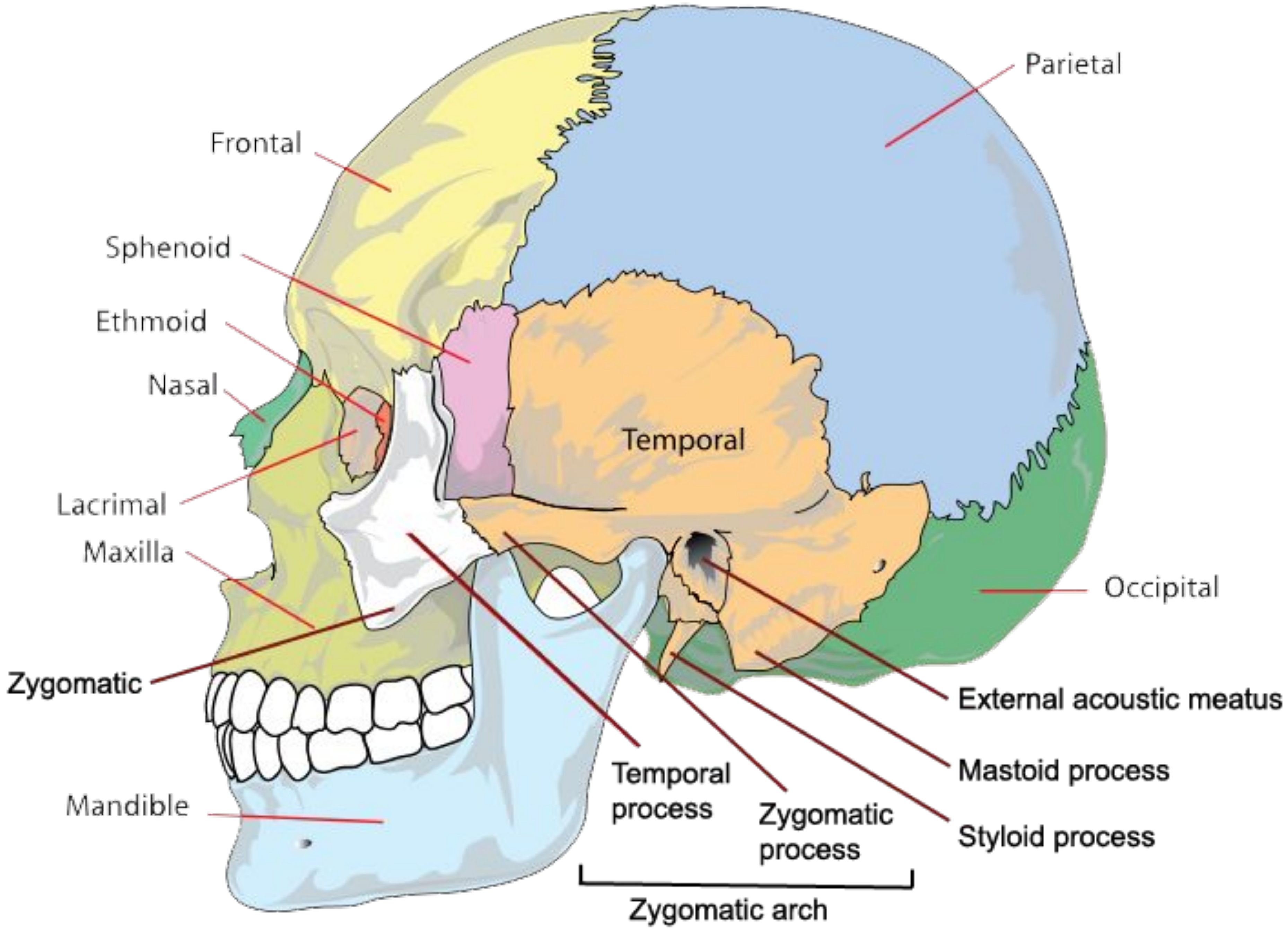
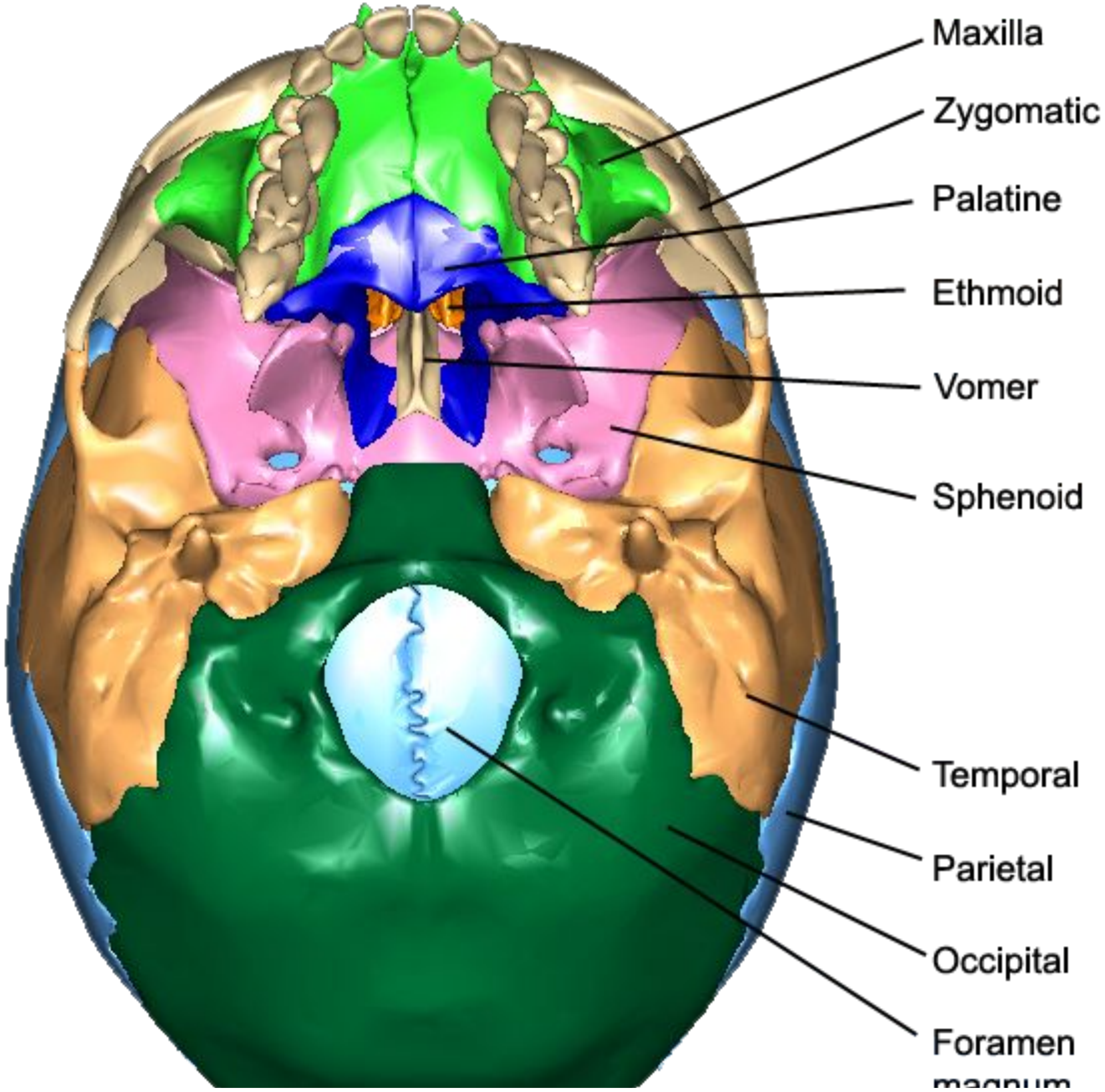


Figure 1: Sagittal section through the temporomandibular joint **(courtesy of Gray' anatomy)**

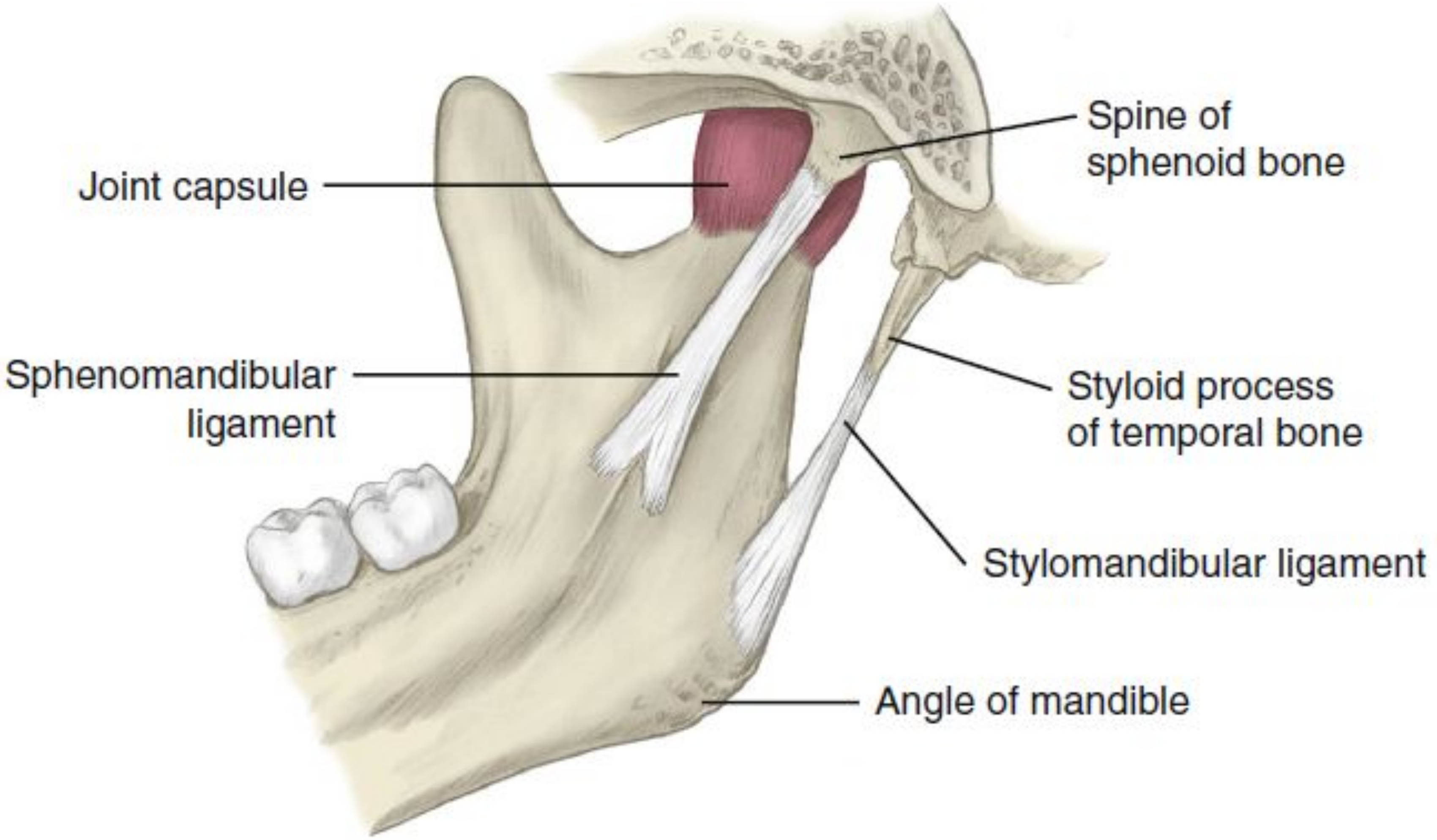

Frontal
Nasal
Parietal
Temporal
Sphenoid
Lacrimal
Zygomatic
Ethmoid
Vomer
Maxilla
Mandible
Supraorbital foramen
Infraorbital foramen
Mental foramen


Parietal
Frontal
Sphenoid
Ethmoid
Nasal
Temporal
Lacrimal
Maxilla
Occipital
Zygomatic
External acoustic meatus
Mastoid process
Temporal process
Zygomatic process
Styloid process
Mandible
Zygomatic arch

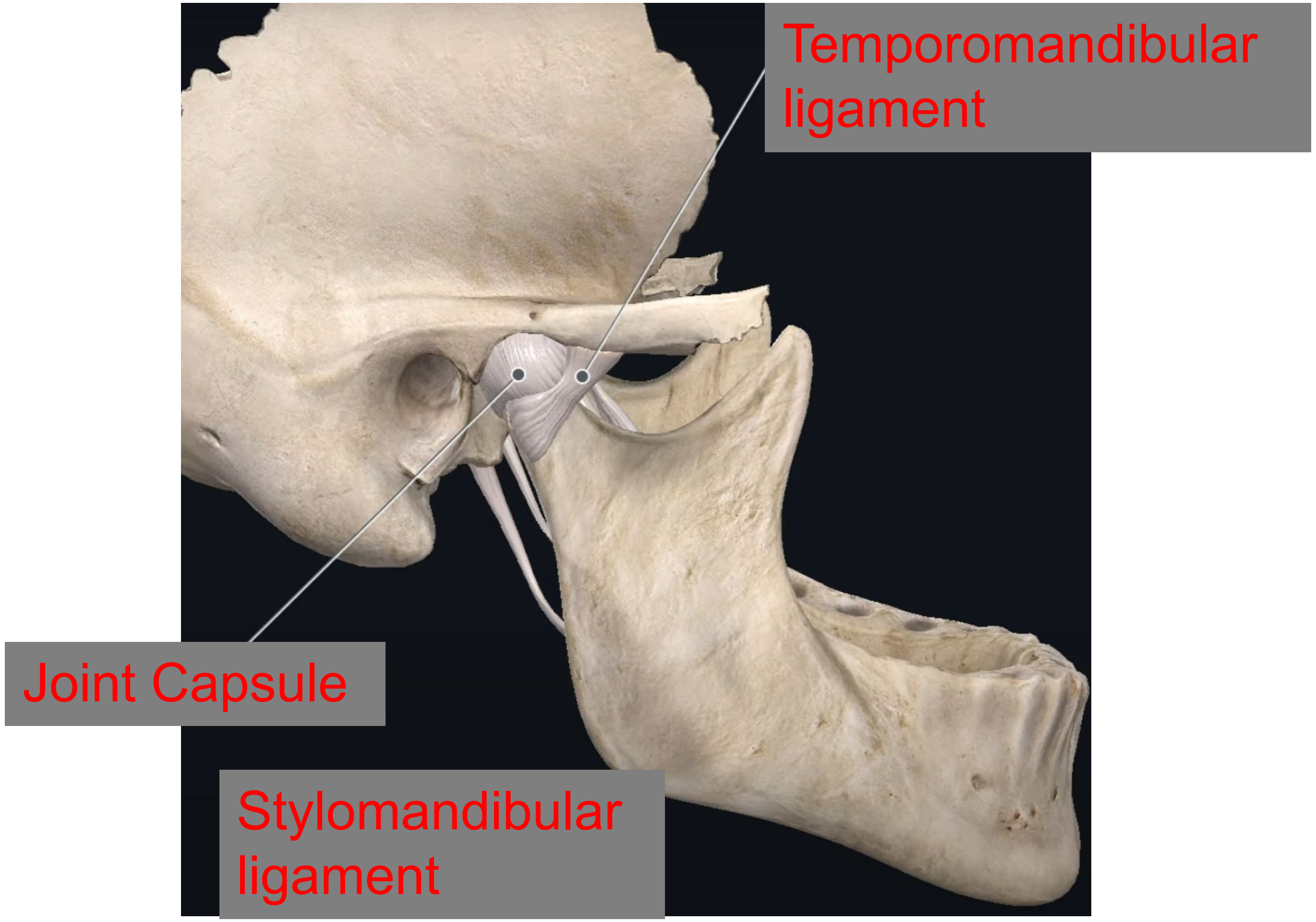

Maxilla
Zygomatic
Palatine
Ethmoid
Vomer
Sphenoid
Temporal
Parietal
Occipital
Foramen

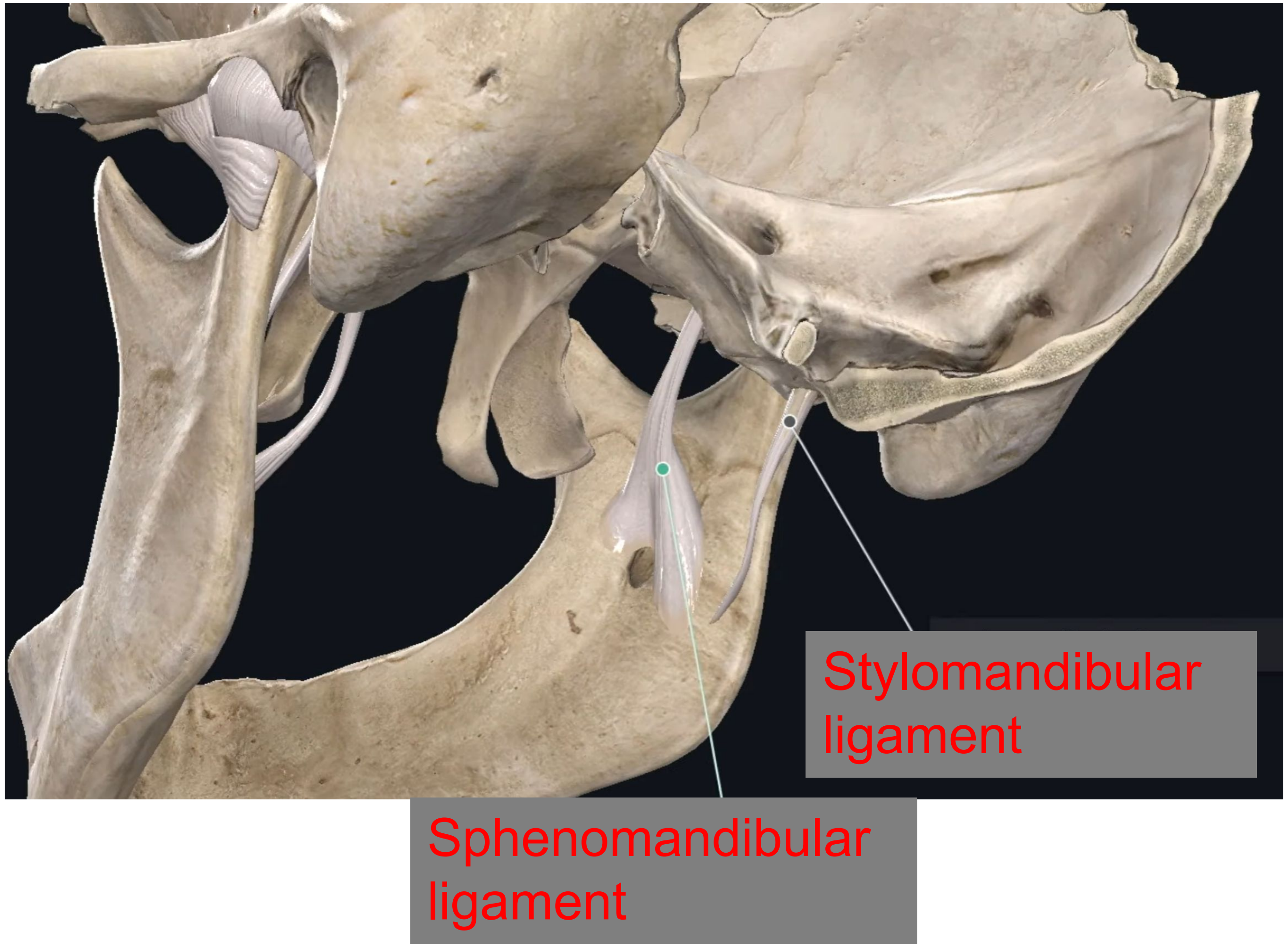
Q1: Enumerate ligaments associated with TMJ, along with their origin, insertion & functions.

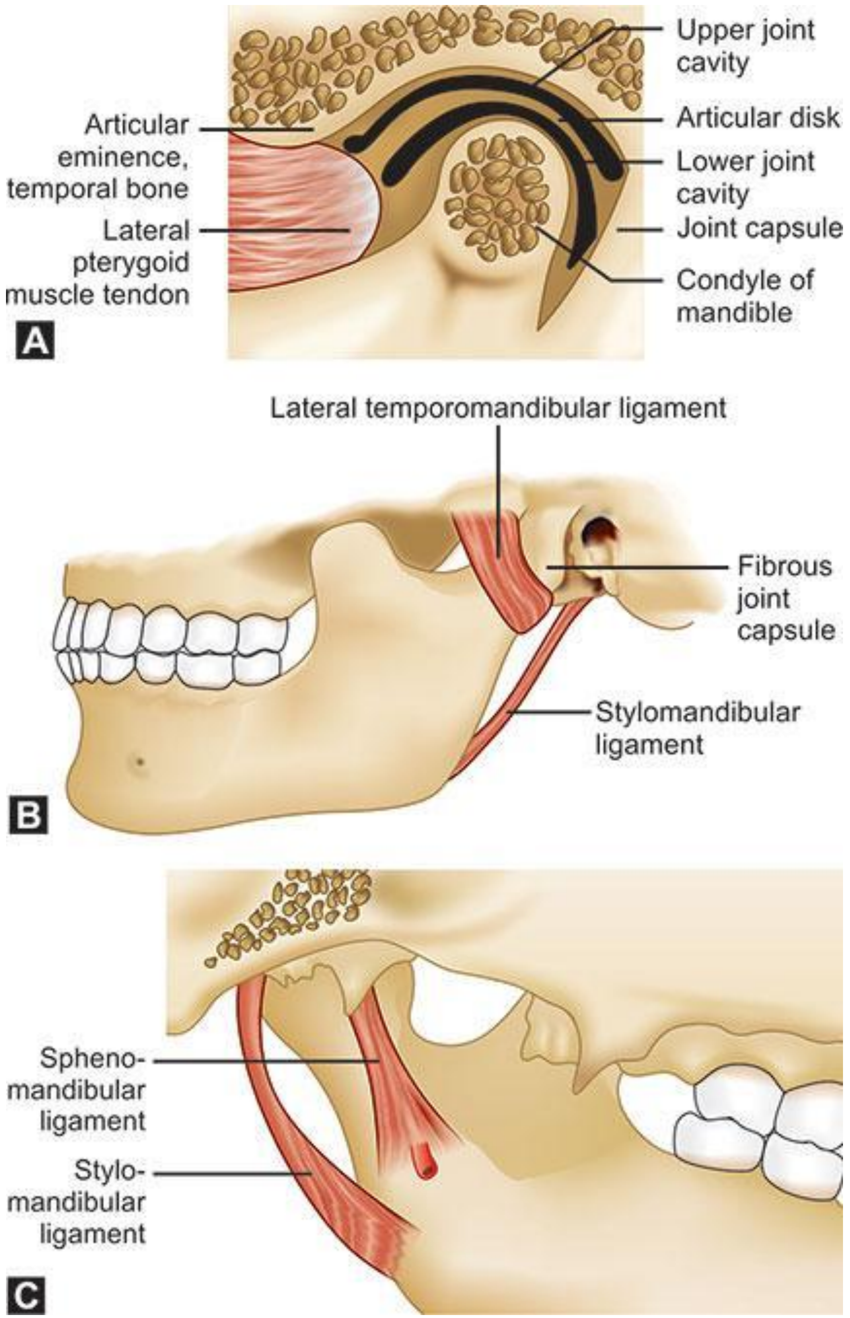
	LATERAL LIGAMENT	ACCESSORY LIGAMENTS				ADDITIONAL LIGAMENT
LIGAMENT NAME	Temporomandibular ligament -	Sphenomandibular ligament	stylomandibular ligament	discomalleolar ligament	pterygomandibular raphe	retinacular ligament
ORIGIN	lateral surface of the articular eminence of the temporal bone, articular tubercle	spine of the sphenoid bone	top of the styloid process of the temporal bone and from the stylohyoid ligament.	medial aspect of the posterosuperior part of the articular disc and capsule	pterygoid hamulus	articular eminernce
INSERTION	posterior surface of the condyle.	lingula near the mandibular foramen	Angle of mandible.	neck and anterior process of the malleus ear ossicle	posterior end of the mylohyoid line in the retromolar region of the mandible.	fascia overlying the masseter muscle at the angle of the mandible.
FUNCTIONS	Strengthens joint capsule & Supports joint, by restricting backward & inferior mandibular movements & resists dislocation during functional movements	Remnant of the perichondrium of Meckel's cartilage. Only ligament with significant influence upon mandibular movements.	It is a reinforced lamina of the deep cervical fascia	tense the synovial membrane during TMJ movement & cause movement of the malleus by tensing.	buccinator and superior constrictor muscles arise from it.	function in maintaining blood circulation during masticatory movements.

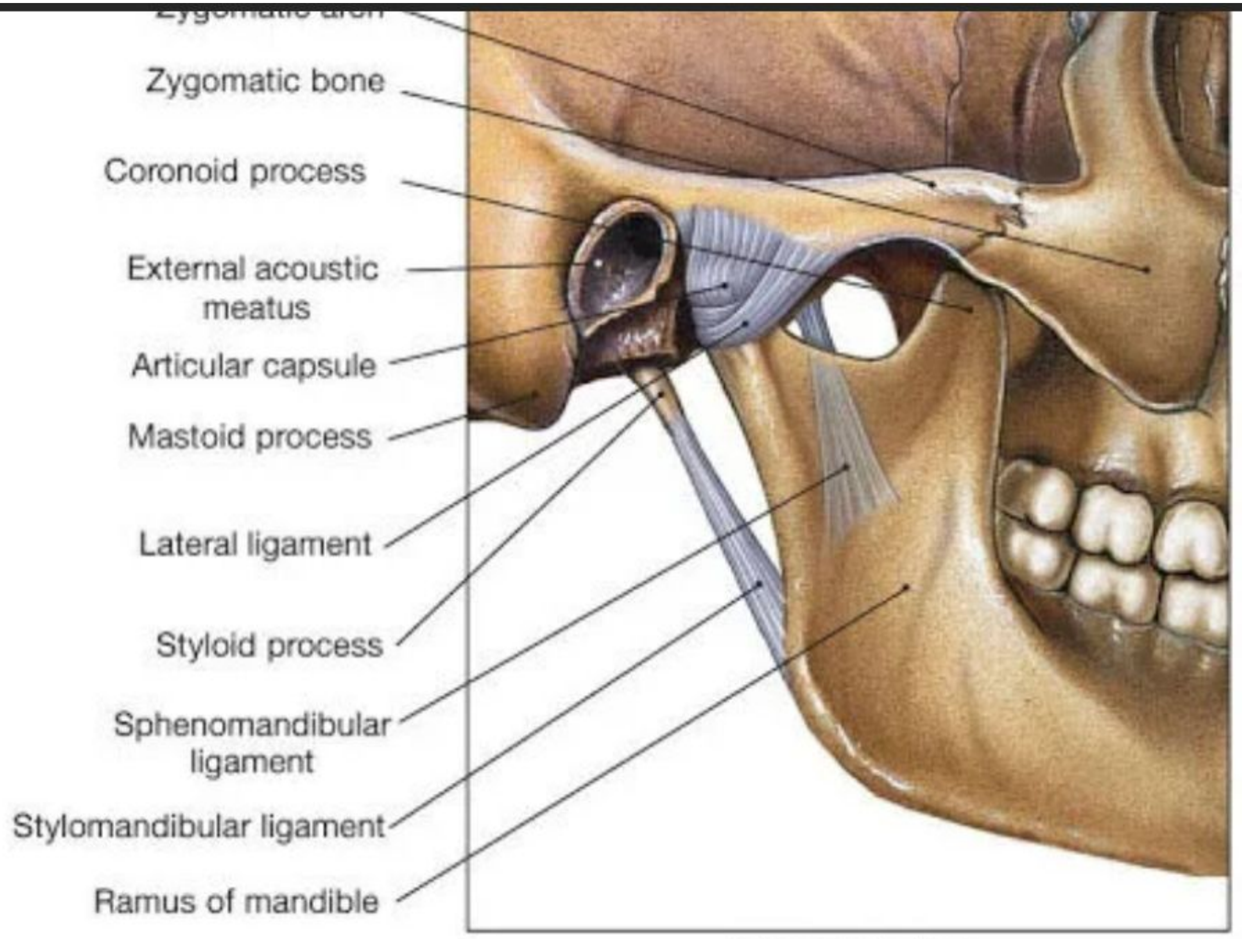


19-5 Joint capsule of the temporomandibular joint. (From Fehrenbach MJ, Herring SW: *Illus-*
natomy of the head and neck, ed 4, St Louis, 2012, Saunders/Elsevier.)


Temporomandibular ligament
Joint Capsule
Stylomandibular ligament


Stylomandibular
ligament
Sphenomandibular
ligament

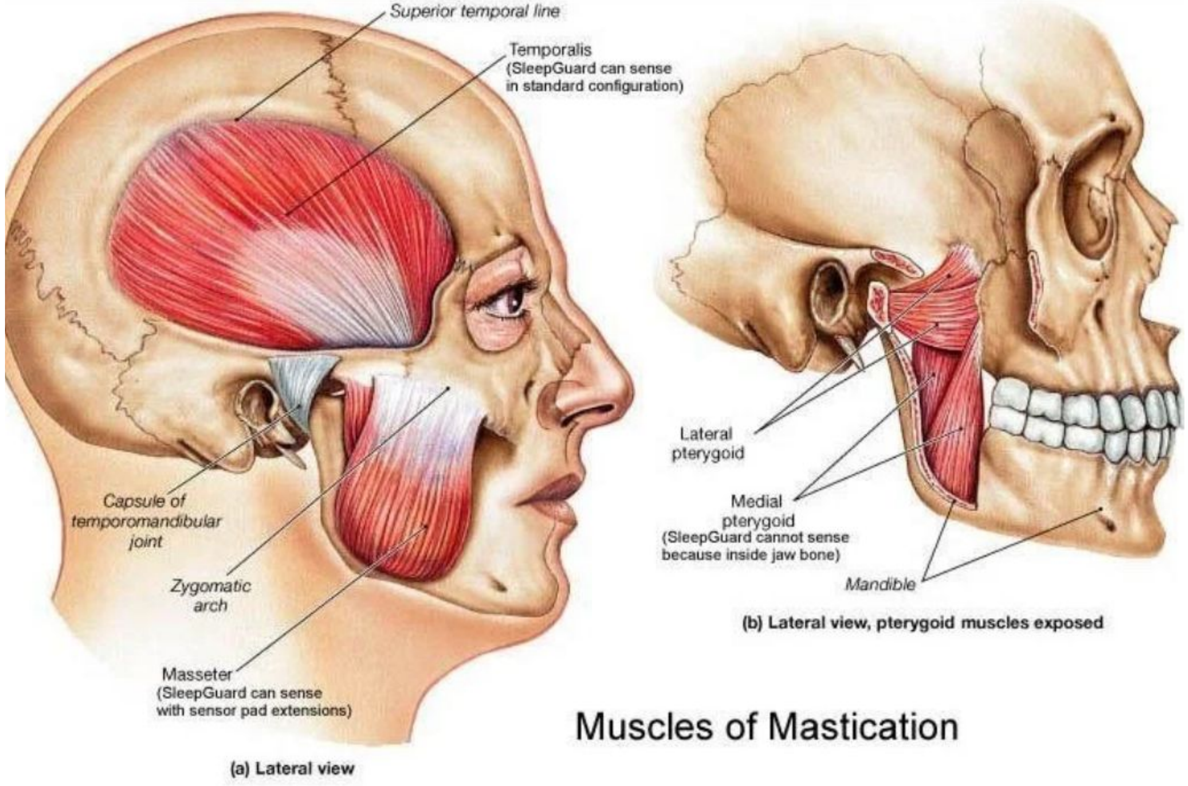




(a) Lateral view

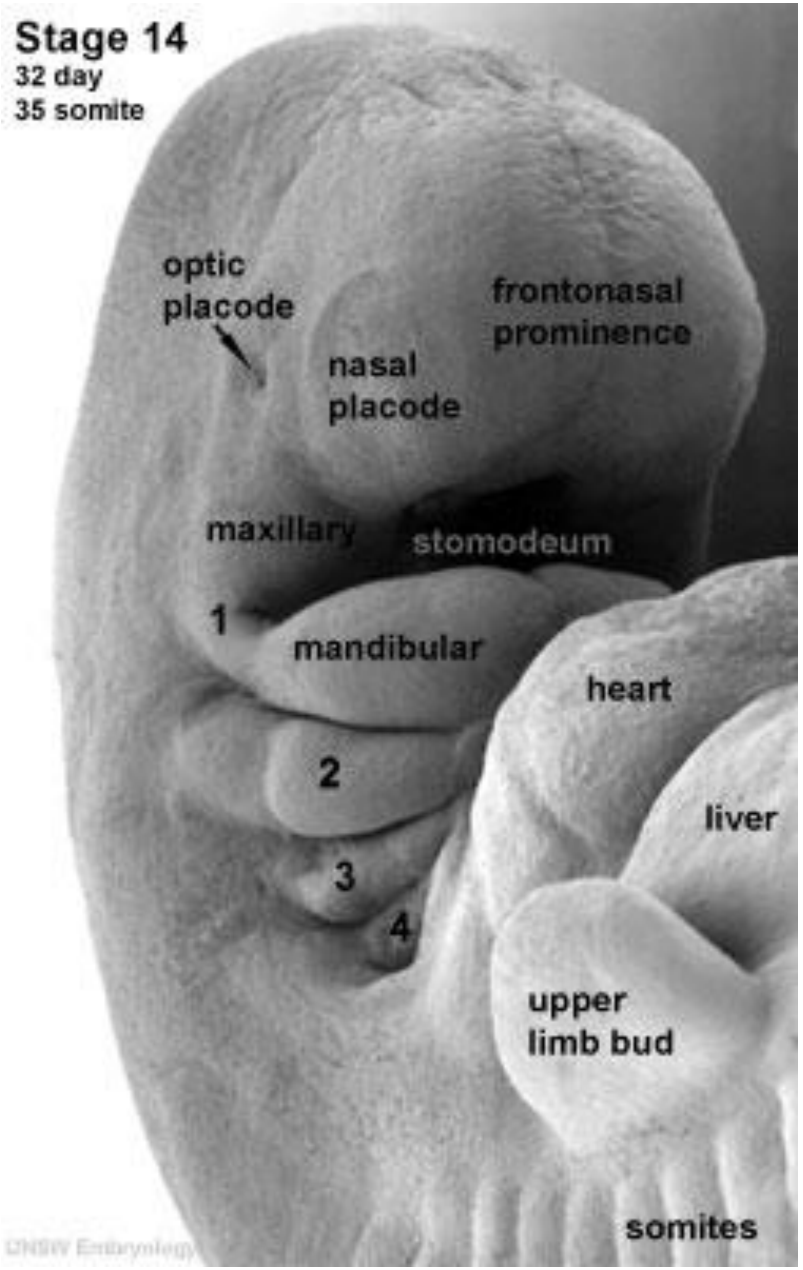
MUSCLES OF MASTICATION

INTRODUCTION



Traditionally four powerful muscles –

Masseter, Temporalis, Medial Pterygoid & Lateral pterygoid are called the "**Muscles of Mastication**"

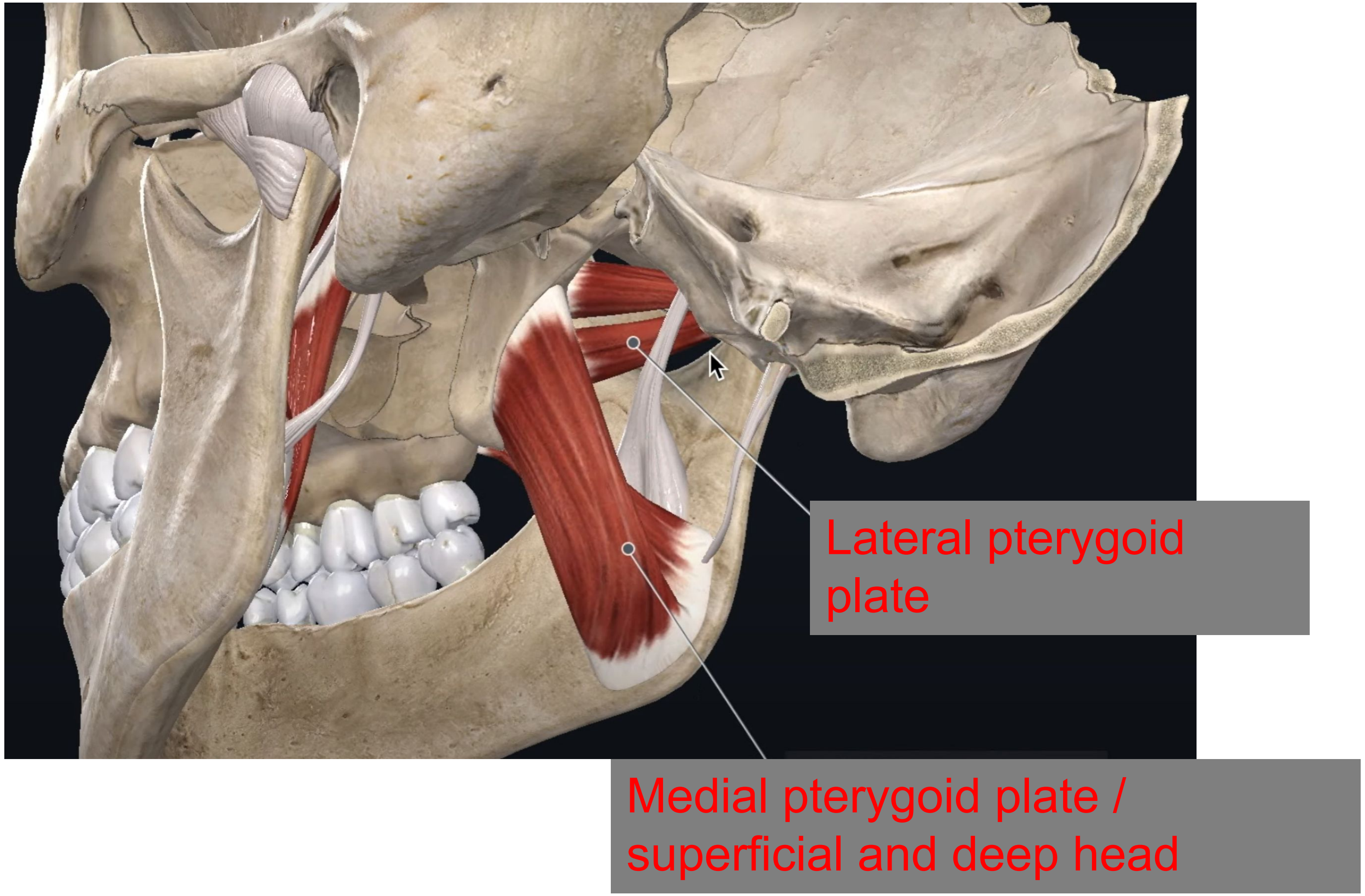


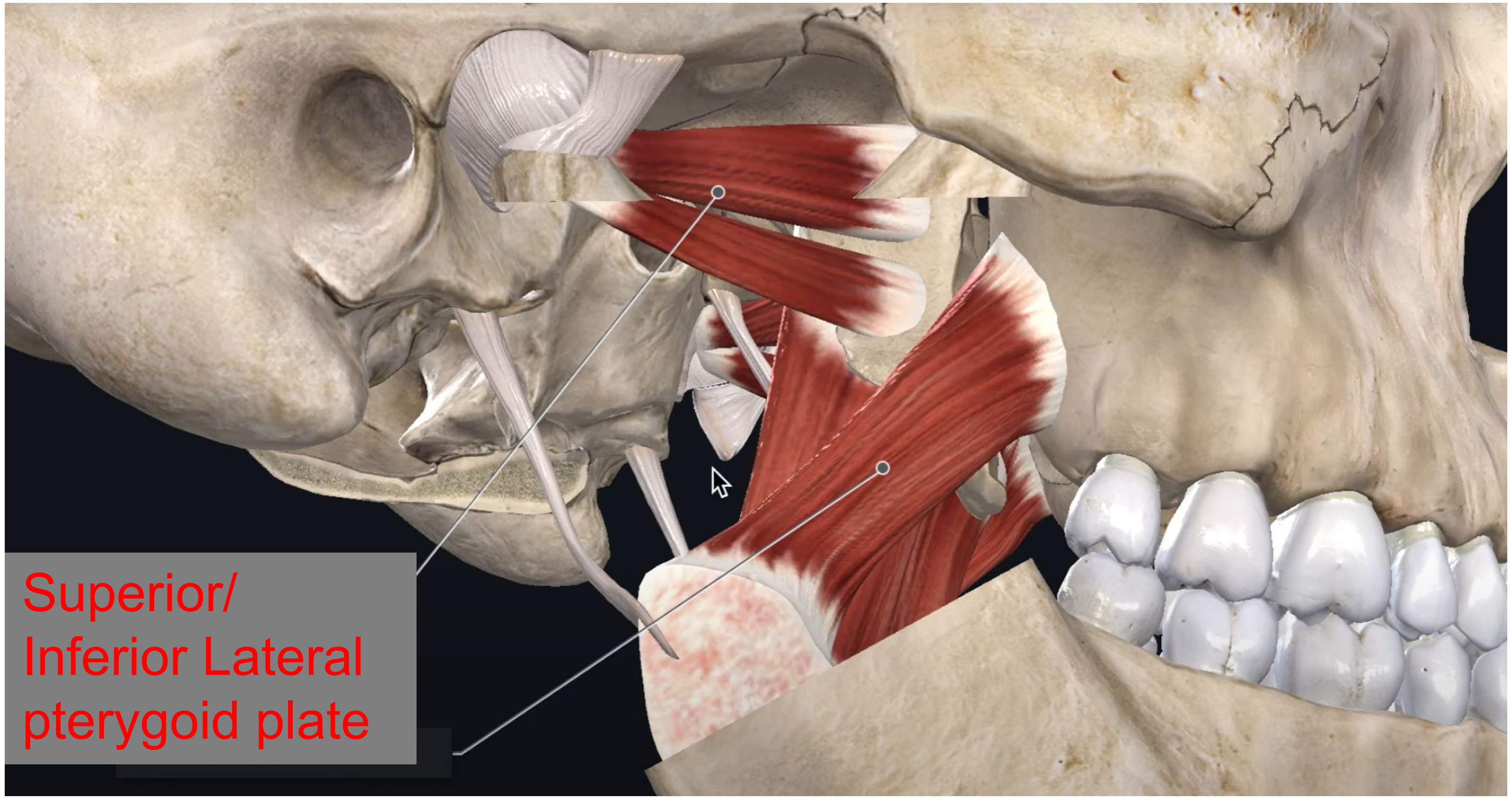
DEVELOPMENT

Muscles of mastication develops from the mesoderm of the first brachial arch that is also called the **Mandibular Arch**

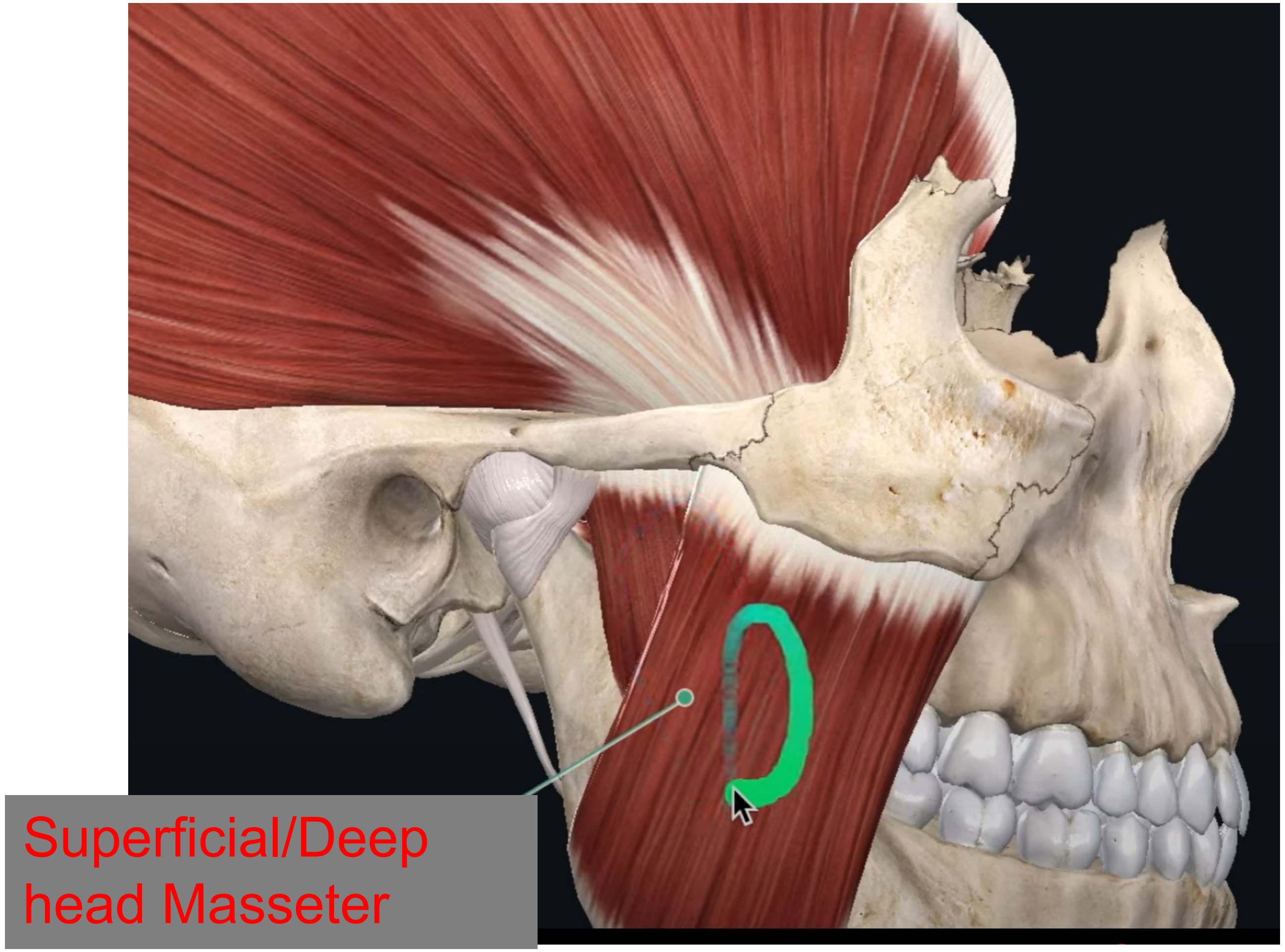
Q1: Enumerate mucles of mastication, along with their origin, insertion & functions.

MUSLE NAME	Masseter	Temporalis -- Largest muscle	Lateral pterygoid	Medial pterygoid
ORIGIN	SUPERFICIAL HEAD: zygomatic process of the maxilla and fromanterior two-thirds of the lower border of the zygomatic arch. DEEP HEAD: deep surface of the zygomatic arch	the floor of the temporal fossa on the lateral surface of the skull and from the overlying temporal fascia.	SUPERIOR (UPPER) HEAD : (smaller) & arises from the infratemporal surface of the greater wing of the sphenoid Bone THE INFERIOR (LOWER) HEAD: forms the bulk of the muscle and takes origin from the lateral surface of the lateral pterygoid plate of the sphenoid bone	DEEP HEAD: medial surface of the lateral pterygoid plate of the sphenoid bone SUPERFICIAL HEAD: maxillary tuberosity , & adjacent process of palatine bone
INSERTION	SUPERFICIAL HEAD: lower half of the lateral surface of the mandibular ramus. DEEP HEAD: upper half of the lateral surface of the ramus, particularly over the coronoid process.	apex, the anterior and posterior borders and the medial surface of the coronoid process of the mandible as far as the third molar tooth.	disc/capsule complex as well as the condyle. neck of the mandibular condyle at the fovea and into the capsule of the joint.	Medial aspect of angle of mandible
FUNCTIONS	Elevation of mandible	The anterior (vertical) part elevates the mandible, while the posterior (horizontal) part retracts the protruded mandible.	INFERIOR HEAD: Protrusion, depression & larteral excusion SUPERIOR HEAD: controlled retrusion,	Elevation of mandible. Also help in lateral & protrusive movements.


Lateral pterygoid plate
Medial pterygoid plate / superficial and deep head


Superior/
Inferior Lateral
pterygoid plate


Temporalis


Superficial/Deep
head Masseter

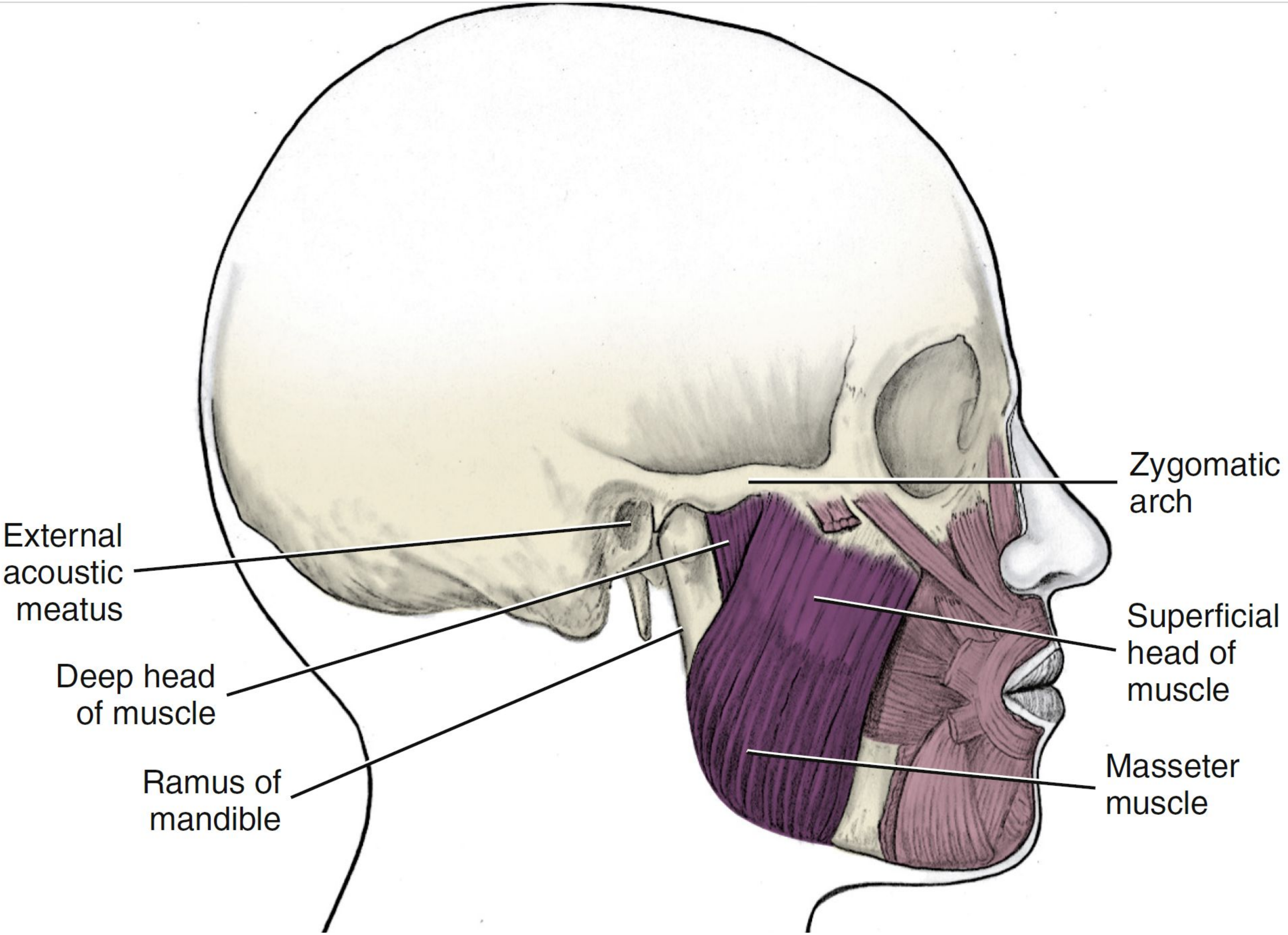


FIGURE 19-8 Muscles of mastication. **A,** Masseter muscle. (**A** and **B,** From Fehrenbach MJ, Herring SW: *Illustrated anatomy of the head and neck,* ed 4, St Louis, 2012, Saunders/ Elsevier.)

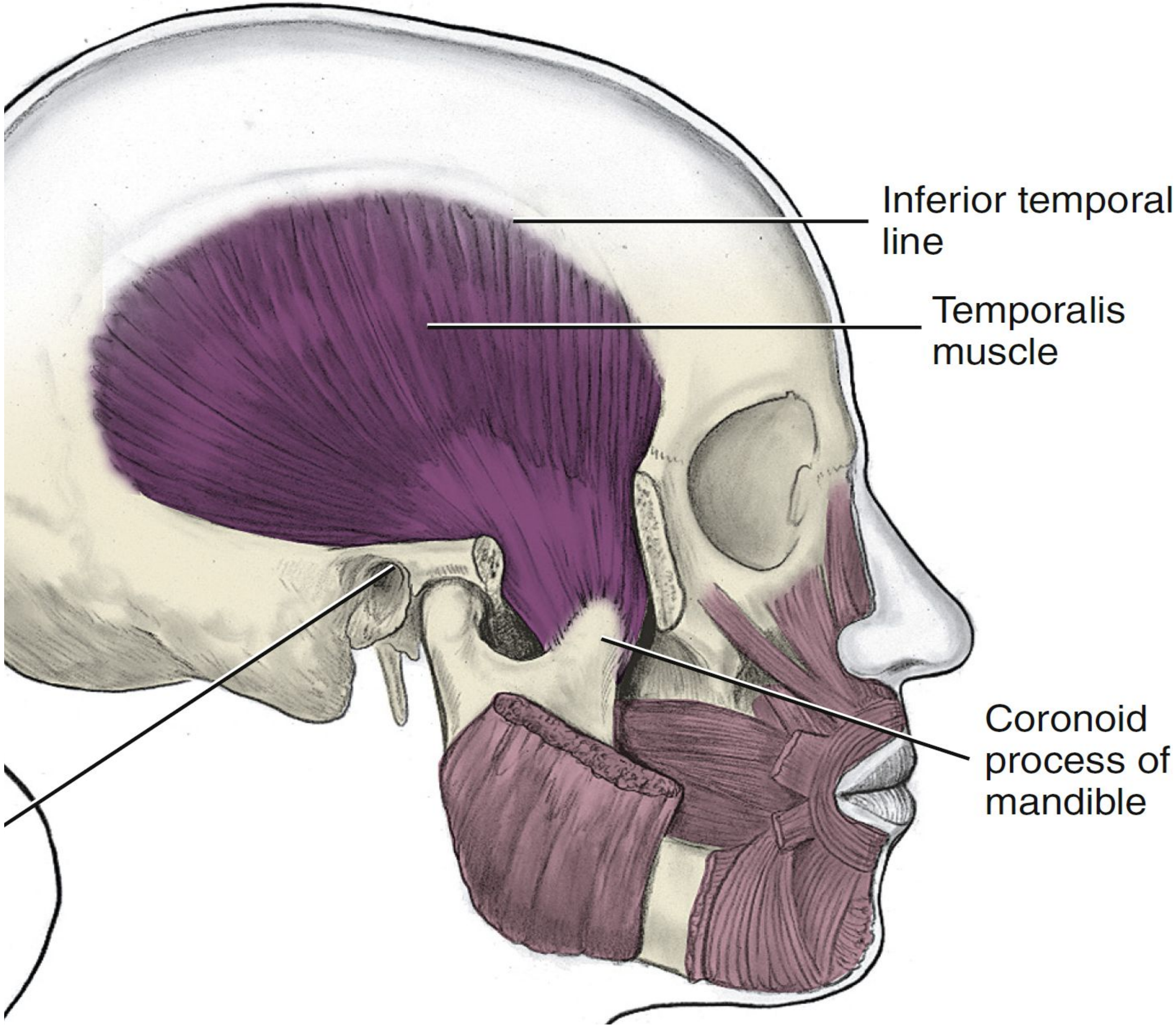


FIGURE 19-8 Muscles of mastication. Temporalis muscle. (**A** and **B,** From Fehrenbach MJ, Herring SW: ***Illustrated anatomy of the head and neck,*** ed 4, St Louis, 2012, Saunders/ Elsevier.)

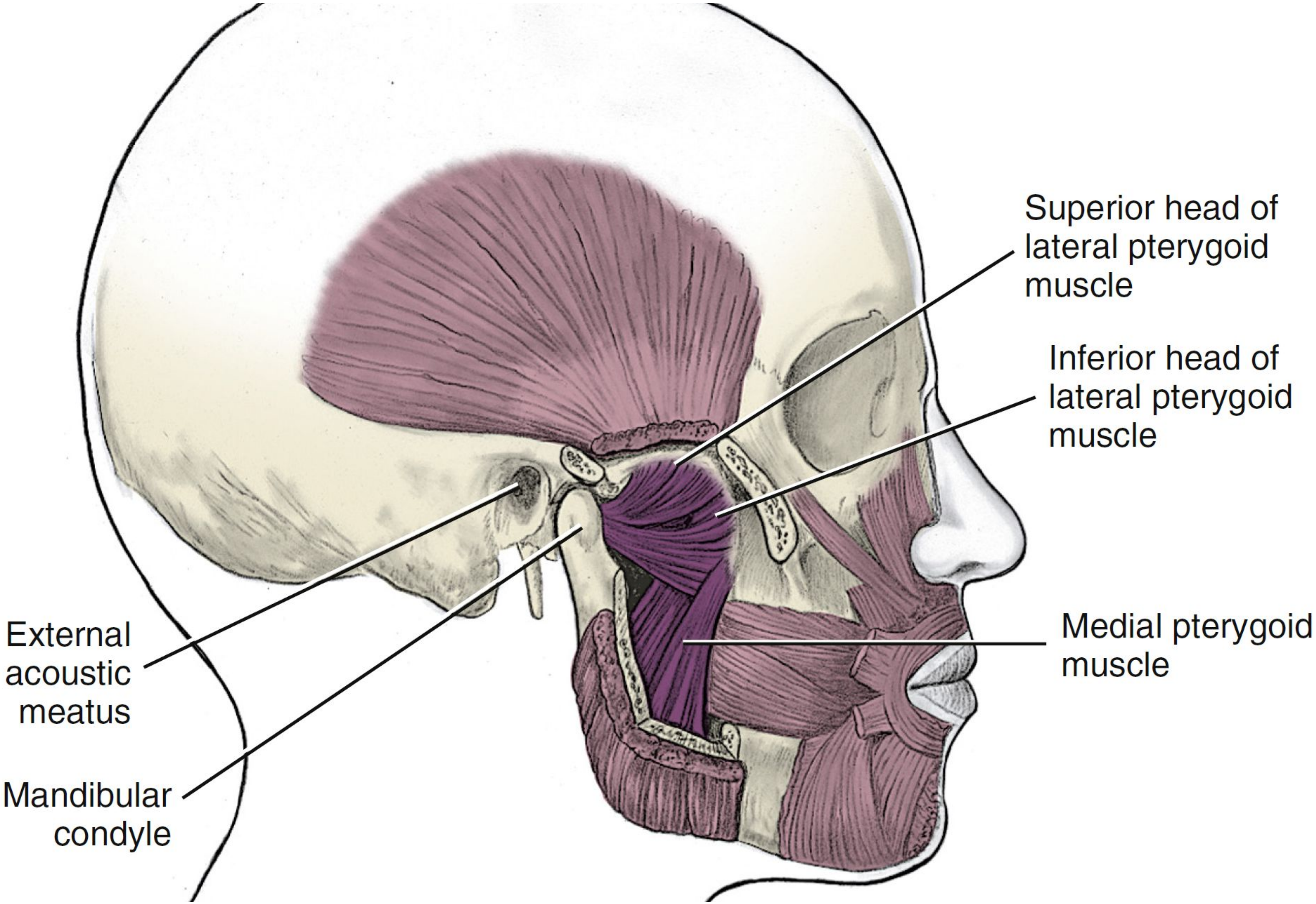
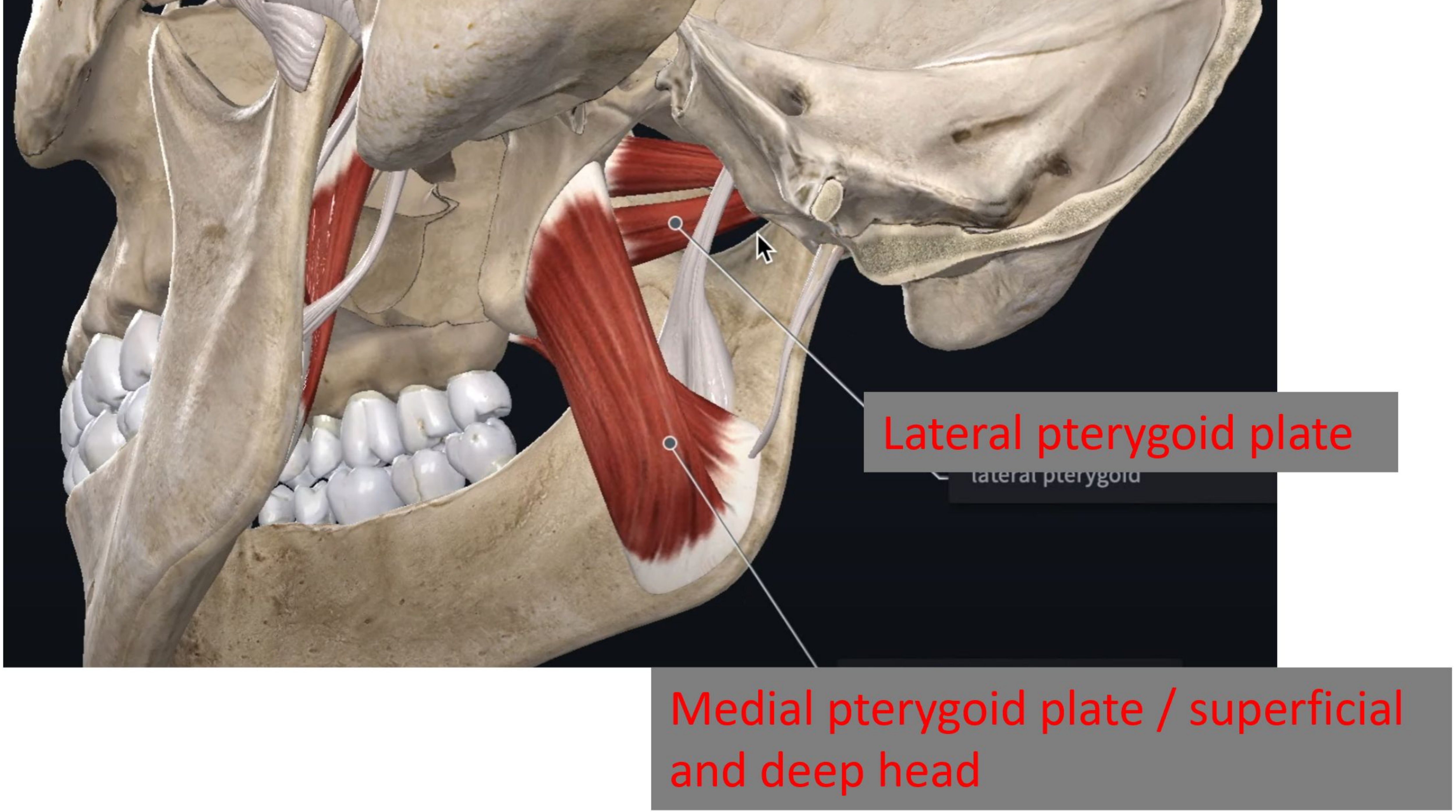


FIGURE 19-8, cont'd Muscles of mastication. **C,** Medial and lateral pterygoid muscles.


Lateral pterygoid plate
lateral pterygoid
Medial pterygoid plate / superficial and deep head